

Supplemental Digital Content 1.

“Prenatal and Childhood Exposure to Air Pollution and Traffic and the Risk of Liver Injury in European Children: The HELIX Project”

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eTable 1. Effective degrees of freedom (EDF)^a and P values^b of fitted Generalized Additive Models presented in eFigures 1 and 1-8 that model liver injury biomarkers as smooth functions of exposure.^c

<i>Exposure</i>	ALT		AST		GGT		CK-18	
	<i>EDF</i>	<i>P values</i>	<i>EDF</i>	<i>P values</i>	<i>EDF</i>	<i>P values</i>	<i>EDF</i>	<i>P values</i>
NO2: Pregnancy	1.000	0.99	1.004	0.37	1.003	0.79	1.005	0.39
NO2: Trimester 1	1.008	0.38	1.002	0.77	1.003	0.10	1.005	0.88
NO2: Trimester 2	1.000	0.52	1.003	0.44	1.001	0.34	1.003	0.58
NO2: Trimester 3	1.572	0.56	1.676	0.24	1.002	0.44	1.001	0.05
NO2: Childhood, home	2.357 ^d	0.32	1.001	0.25	1.001	0.16	1.004	0.75
NO2: Childhood, school	1.001	0.19	1.026	0.71	1.009	0.25	1.003	0.77
PM10: Pregnancy	1.002	0.96	1.003	0.80	1.002	0.77	1.004	0.27
PM10: Trimester 1	1.000	0.66	4.697 ^d	0.002	1.449	0.72	1.201	0.81
PM10: Trimester 2	1.000	0.44	1.059	0.13	1.023	0.45	1.003	0.33
PM10: Trimester 3	1.004	0.80	1.002	0.32	1.002	0.85	1.003	0.13
PM10: Childhood, home	1.475	0.30	1.003	0.05	2.079	0.34	1.000	0.44
PM10: Childhood, school	1.001	0.99	1.002	0.47	2.155	0.34	1.003	0.36
PM2.5: Pregnancy	1.160	0.82	1.003	0.84	1.003	0.16	1.004	0.95
PM2.5: Trimester 1	1.000	0.58	1.002	0.76	1.001	0.80	1.003	0.94
PM2.5: Trimester 2	1.001	0.81	1.002	0.86	1.003	0.40	1.003	0.44
PM2.5: Trimester 3	1.002	0.28	1.003	0.99	1.002	0.04	1.003	0.46
PM2.5: Childhood, home	1.001	0.72	1.006	0.28	1.005	0.92	1.997	0.31
PM2.5: Childhood, school	2.337	0.52	1.003	0.33	1.006	0.85	1.003	0.72
Inverse Distance: Pregnancy	1.000	0.67	3.402 ^d	0.01	1.002	0.44	1.001	0.95
Inverse Distance: Childhood, home	1.987	0.42	1.002	0.26	1.010	0.86	1.005	0.18
Inverse Distance: Childhood, school	1.001	0.31	1.003	0.36	1.004	0.79	1.002	0.35
Traffic Load: Pregnancy	2.284 ^d	0.20	1.375	0.68	1.001	0.59	1.005	0.31
Traffic Load: Childhood, home	1.995	0.39	2.405	0.41	1.993	0.39	1.006	0.35
Traffic Load: Childhood, school	1.583	0.22	1.013	0.82	1.001	0.94	1.002	0.91
Traffic Density: Pregnancy	1.003	0.36	1.004	0.46	1.434	0.66	1.004	0.18
Traffic Density: Childhood, home	1.021	0.75	1.002	0.03	1.011	0.63	1.004	0.74
Traffic Density: Childhood, school	1.005	0.15	1.029	0.52	1.057	0.77	1.003	0.65
Indoor NO2: Childhood, home	1.001	0.46	1.004	0.70	1.001	0.88	2.417	0.19
Indoor PM2.5: Childhood, home	1.164	0.90	1.214	0.09	2.464 ^d	0.16	1.002	0.46
Indoor Benzene: Childhood, home	2.146	0.18	1.652	0.08	1.006	0.27	1.003	0.63
Indoor BTEX: Childhood, home	1.694	0.37	1.002	0.55	1.280	0.77	1.004	0.37

Note: ALT, alanine aminotransferase ; AST, aspartate aminotransferase; BTEX, sum of benzene, toluene, ethylbenzene, and xylene; CK-18, cytokeratin-18; GGT, gamma-glutamyltransferase;

NO2, nitrogen dioxide; PM2.5, particulate matter <2.5 µm; PM10, particulate matter <10 µm.

^a EDF for the smooth term; a value of 1 indicates a linear effect, higher values indicate a higher degree of non-linearity.

^b P values shown are unadjusted for multiple comparisons. No P values were statistically significant after adjustment for multiple comparisons.

^c Models adjusted for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy.

^d Statistically significant evidence of non-linearity based on Wald test of smooth term in addition to linear term

eTable 2. Summary of results from models for alanine aminotransferase (ALT) with tertiles of exposure.^a

Exposure	Terile 1 (Ref)	Tertile 2			Tertile 3		
	Range ^b	Range ^b	beta ± SE	P value ^c	Range ^b	beta ± SE	P value ^c
NO2: Pregnancy	[3.1,3.37]	(3.37,3.59]	-0.84 ± 0.51	0.10	(3.59,4.67]	-1.21 ± 0.62	0.05
NO2: Trimester 1	[2.96,3.4]	(3.4,3.65]	-0.36 ± 0.48	0.46	(3.65,4.74]	-0.42 ± 0.59	0.48
NO2: Trimester 2	[2.9,3.29]	(3.29,3.57]	-0.03 ± 0.49	0.95	(3.57,4.65]	-0.38 ± 0.59	0.52
NO2: Trimester 3	[2.91,3.32]	(3.32,3.57]	0.33 ± 0.49	0.49	(3.57,4.79]	0.23 ± 0.60	0.70
NO2: Childhood, home	[2.61,3.21]	(3.21,3.66]	-1.18 ± 0.58	0.04	(3.66,4.56]	-0.63 ± 0.76	0.41
NO2: Childhood, school	[2.66,3.26]	(3.26,3.69]	-0.17 ± 0.58	0.77	(3.69,4.56]	0.13 ± 0.76	0.86
PM10: Pregnancy	[8.07,17.6]	(17.6,26]	0.27 ± 0.63	0.67	(26,47.7]	0.00 ± 0.86	1.00
PM10: Trimester 1	[6.76,17.9]	(17.9,26.1]	-0.78 ± 0.60	0.20	(26.1,47.6]	-0.86 ± 0.77	0.26
PM10: Trimester 2	[6.61,17.6]	(17.6,25.9]	-0.16 ± 0.60	0.79	(25.9,49]	-0.59 ± 0.78	0.45
PM10: Trimester 3	[7.39,16.5]	(16.5,24.9]	-0.28 ± 0.58	0.63	(24.9,55.4]	-0.18 ± 0.77	0.82
PM10: Childhood, home	[11.2,22.1]	(22.1,29.9]	0.78 ± 0.67	0.24	(29.9,47.7]	1.11 ± 0.87	0.20
PM10: Childhood, school	[11.2,21.3]	(21.3,30.5]	0.90 ± 0.70	0.20	(30.5,49.8]	0.85 ± 0.92	0.36
PM2.5: Pregnancy	[6.96,13.7]	(13.7,16]	0.54 ± 0.60	0.94	(16,30.6]	-0.01 ± 0.57	0.98
PM2.5: Trimester 1	[5.29,13]	(13,16.4]	-0.32 ± 0.45	0.48	(16.4,28.2]	-0.20 ± 0.50	0.69
PM2.5: Trimester 2	[6.36,12.5]	(12.5,15.8]	0.41 ± 0.45	0.36	(15.8,37.1]	0.06 ± 0.49	0.90
PM2.5: Trimester 3	[7.01,12.3]	(12.3,15.5]	-0.29 ± 0.44	0.50	(15.5,32]	0.46 ± 0.47	0.34
PM2.5: Childhood, home	[4.82,12.8]	(12.8,14.4]	0.54 ± 0.69	0.43	(14.4,22]	0.24 ± 0.73	0.75
PM2.5: Childhood, school	[4.93,12.8]	(12.8,14.4]	0.68 ± 0.70	0.33	(14.4,21.6]	0.33 ± 0.76	0.66
Inverse Distance: Pregnancy	[-3.97,-3.08]	(-3.08,-2.33]	0.27 ± 0.46	0.56	(-2.33,2.19]	0.45 ± 0.57	0.43
Inverse Distance: Childhood, home	[-4.97,-4.11]	(-4.11,-3.05]	0.17 ± 0.45	0.70	(-3.05,4.57]	0.79 ± 0.51	0.12
Inverse Distance: Childhood, school	[-4.87,-4.05]	(-4.05,-3.44]	1.12 ± 0.46	0.01	(-3.44,0.973]	0.42 ± 0.50	0.39
Traffic Load: Pregnancy	0	(0,99.3]	1.01 ± 0.47	0.03	(99.3,294]	0.65 ± 0.44	0.15
Traffic Load: Childhood, home	0	(0,97]	0.59 ± 0.49	0.23	(97,294]	0.52 ± 0.46	0.26
Traffic Load: Childhood, school	[0,6.33]	(6.33,96.7]	1.13 ± 0.48	0.02	(96.7,276]	0.88 ± 0.46	0.06
Traffic Density: Pregnancy	[0,7.94]	(7.94,13.9]	-0.32 ± 0.49	0.52	(13.9,39.3]	-0.32 ± 0.46	0.49
Traffic Density: Childhood, home	[0,10]	(10,19.8]	0.07 ± 0.48	0.88	(19.8,49.3]	-0.27 ± 0.59	0.64
Traffic Density: Childhood, school	[0,10.5]	(10.5,19.6]	0.60 ± 0.46	0.20	(19.6,51.2]	-0.63 ± 0.56	0.26
Indoor NO2: Childhood, home	[3.16,3.67]	(3.67,4.46]	-0.08 ± 0.46	0.86	(4.46,6.96]	0.23 ± 0.64	0.72
Indoor PM2.5: Childhood, home	[2.53,2.78]	(2.78,2.98]	-0.69 ± 0.44	0.11	(2.98,5.28]	-0.1 ± 0.48	0.83
Indoor Benzene: Childhood, home	[0.737,1.06]	(1.06,1.26]	0.07 ± 0.44	0.88	(1.26,3.16]	-0.15 ± 0.47	0.76
Indoor BTEX: Childhood, home	[3.03,3.41]	(3.41,3.71]	-0.25 ± 0.45	0.58	(3.71,5.04]	0.03 ± 0.54	0.96

Note: ALT, alanine aminotransferase; BTEX, sum of benzene, toluene, ethylbenzene, and xylene; NO2, nitrogen dioxide; PM2.5, particulate matter <2.5 µm; PM10, particulate matter <10 µm.

^a Models adjusted for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy.

^b Ranges shown are for transformed exposure values, when exposure was transformed.

^c P values shown are unadjusted for multiple comparisons. No P values were statistically significant after adjustment for multiple comparisons.

eTable 3. Summary of results from models for aspartate aminotransferase (AST) with tertiles of exposure. ^a							
	Terile 1 (Ref)	Tertile 2			Tertile 3		
Exposure	<i>Range^b</i>	<i>Range^b</i>	<i>beta ± SE</i>	<i>P value^c</i>	<i>Range^b</i>	<i>beta ± SE</i>	<i>P value^c</i>
NO2: Pregnancy	[3.1,3.37]	(3.37,3.59]	0.83 ± 0.64	0.19	(3.59,4.67]	0.26 ± 0.77	0.74
NO2: Trimester 1	[2.96,3.4]	(3.4,3.65]	0.54 ± 0.60	0.37	(3.65,4.74]	0.35 ± 0.73	0.63
NO2: Trimester 2	[2.9,3.29]	(3.29,3.57]	0.53 ± 0.61	0.38	(3.57,4.65]	0.73 ± 0.74	0.32
NO2: Trimester 3	[2.91,3.32]	(3.32,3.57]	1.46 ± 0.61	0.02	(3.57,4.79]	1.57 ± 0.75	0.04
NO2: Childhood, home	[2.61,3.21]	(3.21,3.66]	0.56 ± 0.73	0.44	(3.66,4.56]	1.01 ± 0.95	0.29
NO2: Childhood, school	[2.66,3.26]	(3.26,3.69]	0.16 ± 0.72	0.82	(3.69,4.56]	0.33 ± 0.95	0.73
PM10: Pregnancy	[8.07,17.6]	(17.6,26]	-0.14 ± 0.79	0.86	(26,47.7]	-0.79 ± 1.07	0.46
PM10: Trimester 1	[6.76,17.9]	(17.9,26.1]	0.33 ± 0.75	0.66	(26.1,47.6]	0.08 ± 0.97	0.94
PM10: Trimester 2	[6.61,17.6]	(17.6,25.9]	0.10 ± 0.75	0.89	(25.9,49]	-0.69 ± 0.98	0.48
PM10: Trimester 3	[7.39,16.5]	(16.5,24.9]	0.50 ± 0.72	0.49	(24.9,55.4]	0.36 ± 0.96	0.71
PM10: Childhood, home	[11.2,22.1]	(22.1,29.9]	0.96 ± 0.84	0.25	(29.9,47.7]	1.10 ± 1.09	0.31
PM10: Childhood, school	[11.2,21.3]	(21.3,30.5]	1.23 ± 0.87	0.16	(30.5,49.8]	0.93 ± 1.15	0.42
PM2.5: Pregnancy	[6.96,13.7]	(13.7,16]	-0.33 ± 0.62	0.60	(16,30.6]	0.41 ± 0.71	0.57
PM2.5: Trimester 1	[5.29,13]	(13,16.4]	0.29 ± 0.56	0.61	(16.4,28.2]	0.68 ± 0.62	0.27
PM2.5: Trimester 2	[6.36,12.5]	(12.5,15.8]	-0.08 ± 0.56	0.88	(15.8,37.1]	0.07 ± 0.61	0.90
PM2.5: Trimester 3	[7.01,12.3]	(12.3,15.5]	-0.38 ± 0.55	0.49	(15.5,32]	0.24 ± 0.59	0.68
PM2.5: Childhood, home	[4.82,12.8]	(12.8,14.4]	0.22 ± 0.86	0.80	(14.4,22]	0.11 ± 0.92	0.91
PM2.5: Childhood, school	[4.93,12.8]	(12.8,14.4]	-0.12 ± 0.87	0.89	(14.4,21.6]	-0.38 ± 0.95	0.69
Inverse Distance: Pregnancy	[-3.97,-3.08]	(-3.08,-2.33]	0.55 ± 0.58	0.35	(-2.33,2.19]	-0.44 ± 0.72	0.54
Inverse Distance: Childhood, home	[-4.97,-4.11]	(-4.11,-3.05]	-0.03 ± 0.56	0.96	(-3.05,4.57]	0.27 ± 0.63	0.67
Inverse Distance: Childhood, school	[-4.87,-4.05]	(-4.05,-3.44]	-0.41 ± 0.57	0.48	(-3.44,0.973]	-0.48 ± 0.62	0.44
Traffic Load: Pregnancy	0	(0,99.3]	0.49 ± 0.59	0.40	(99.3,294]	-0.08 ± 0.56	0.89
Traffic Load: Childhood, home	0	(0,97]	1.21 ± 0.61	0.05	(97,294]	0.10 ± 0.57	0.87
Traffic Load: Childhood, school	[0,6.33]	(6.33,96.7]	-0.16 ± 0.60	0.79	(96.7,276]	-0.23 ± 0.58	0.69
Traffic Density: Pregnancy	[0,7.94]	(7.94,13.9]	-0.26 ± 0.61	0.67	(13.9,39.3]	-0.52 ± 0.58	0.37
Traffic Density: Childhood, home	[0,10]	(10,19.8]	-1.18 ± 0.60	0.05	(19.8,49.3]	-1.49 ± 0.74	0.04
Traffic Density: Childhood, school	[0,10.5]	(10.5,19.6]	0.36 ± 0.58	0.54	(19.6,51.2]	0.28 ± 0.70	0.69
Indoor NO2: Childhood, home	[3.16,3.67]	(3.67,4.46]	-0.09 ± 0.57	0.88	(4.46,6.96]	-0.40 ± 0.79	0.61
Indoor PM2.5: Childhood, home	[2.53,2.78]	(2.78,2.98]	0.22 ± 0.55	0.68	(2.98,5.28]	-0.21 ± 0.60	0.72
Indoor Benzene: Childhood, home	[0.737,1.06]	(1.06,1.26]	-1.01 ± 0.54	0.06	(1.26,3.16]	-0.84 ± 0.59	0.16
Indoor BTEX: Childhood, home	[3.03,3.41]	(3.41,3.71]	-0.21 ± 0.57	0.71	(3.71,5.04]	0.52 ± 0.67	0.44

Note: AST, aspartate aminotransferase; BTEX, sum of benzene, toluene, ethylbenzene, and xylene; NO2, nitrogen dioxide; PM2.5, particulate matter <2.5 µm; PM10, particulate matter <10 µm.

^a Models adjusted for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy.

^b Ranges shown are for transformed exposure values, when exposure was transformed.

^c P values shown are unadjusted for multiple comparisons. No P values were statistically significant after adjustment for multiple comparisons.

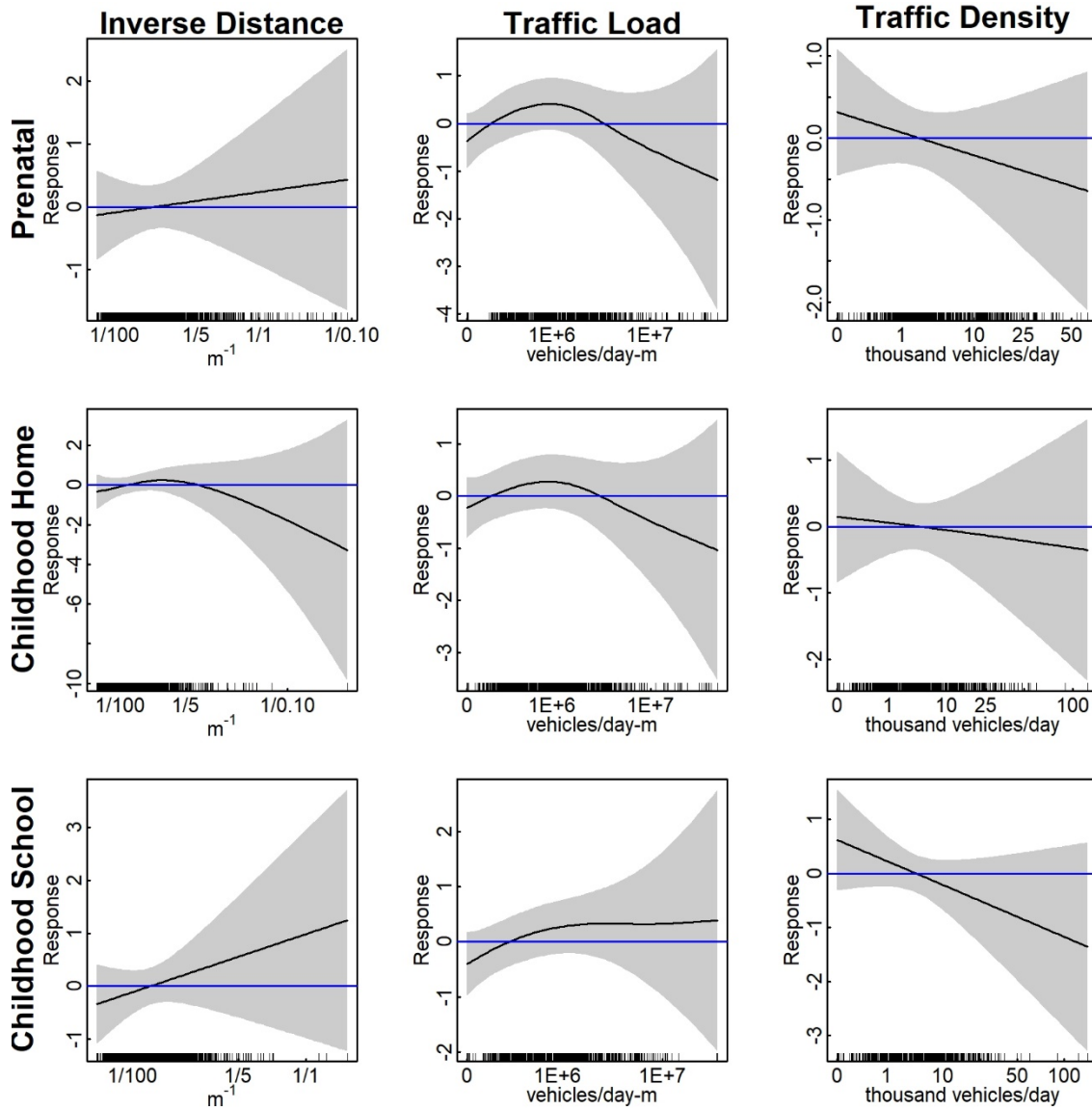
	Tertile 1 (Ref)	Tertile 2			Tertile 3		
Exposure	Range ^b	Range ^b	beta ± SE	P value ^c	Range ^b	beta ± SE	P value ^c
NO2: Pregnancy	[3.1,3.37]	(3.37,3.59]	0.06 ± 0.29	0.85	(3.59,4.67]	-0.13 ± 0.35	0.71
NO2: Trimester 1	[2.96,3.4]	(3.4,3.65]	0.06 ± 0.28	0.84	(3.65,4.74]	-0.30 ± 0.34	0.37
NO2: Trimester 2	[2.9,3.29]	(3.29,3.57]	0.17 ± 0.28	0.55	(3.57,4.65]	0.35 ± 0.34	0.30
NO2: Trimester 3	[2.91,3.32]	(3.32,3.57]	0.20 ± 0.28	0.48	(3.57,4.79]	0.73 ± 0.34	0.03
NO2: Childhood, home	[2.61,3.21]	(3.21,3.66]	0.24 ± 0.33	0.48	(3.66,4.56]	0.35 ± 0.44	0.42
NO2: Childhood, school	[2.66,3.26]	(3.26,3.69]	0.07 ± 0.33	0.84	(3.69,4.56]	0.23 ± 0.43	0.59
PM10: Pregnancy	[8.07,17.6]	(17.6,26]	-0.23 ± 0.36	0.52	(26,47.7]	0.06 ± 0.49	0.90
PM10: Trimester 1	[6.76,17.9]	(17.9,26.1]	-0.06 ± 0.35	0.86	(26.1,47.6]	0.06 ± 0.44	0.88
PM10: Trimester 2	[6.61,17.6]	(17.6,25.9]	-0.33 ± 0.34	0.33	(25.9,49]	-0.51 ± 0.45	0.25
PM10: Trimester 3	[7.39,16.5]	(16.5,24.9]	0.16 ± 0.33	0.63	(24.9,55.4]	0.24 ± 0.44	0.58
PM10: Childhood, home	[11.2,22.1]	(22.1,29.9]	0.81 ± 0.38	0.03	(29.9,47.7]	0.67 ± 0.50	0.18
PM10: Childhood, school	[11.2,21.3]	(21.3,30.5]	0.10 ± 0.40	0.79	(30.5,49.8]	0.42 ± 0.53	0.43
PM2.5: Pregnancy	[6.96,13.7]	(13.7,16]	-0.06 ± 0.28	0.85	(16,30.6]	0.47 ± 0.33	0.15
PM2.5: Trimester 1	[5.29,13]	(13,16.4]	0.05 ± 0.26	0.83	(16.4,28.2]	0.02 ± 0.28	0.94
PM2.5: Trimester 2	[6.36,12.5]	(12.5,15.8]	-0.21 ± 0.26	0.41	(15.8,37.1]	0.38 ± 0.28	0.18
PM2.5: Trimester 3	[7.01,12.3]	(12.3,15.5]	0.08 ± 0.25	0.74	(15.5,32]	0.61 ± 0.27	0.02
PM2.5: Childhood, home	[4.82,12.8]	(12.8,14.4]	0.75 ± 0.40	0.06	(14.4,22]	0.44 ± 0.42	0.30
PM2.5: Childhood, school	[4.93,12.8]	(12.8,14.4]	0.55 ± 0.40	0.17	(14.4,21.6]	-0.01 ± 0.44	0.98
Inverse Distance: Pregnancy	[-3.97,-3.08]	(-3.08,-2.33]	-0.29 ± 0.27	0.28	(-2.33,2.19]	-0.15 ± 0.33	0.65
Inverse Distance: Childhood, home	[-4.97,-4.11]	(-4.11,-3.05]	0.35 ± 0.26	0.17	(-3.05,4.57]	0.12 ± 0.29	0.67
Inverse Distance: Childhood, school	[-4.87,-4.05]	(-4.05,-3.44]	0.39 ± 0.26	0.14	(-3.44,0.973]	-0.13 ± 0.28	0.64
Traffic Load: Pregnancy	0	(0,99.3]	0.24 ± 0.27	0.38	(99.3,294]	-0.03 ± 0.25	0.90
Traffic Load: Childhood, home	0	(0,97]	0.58 ± 0.28	0.04	(97,294]	0.21 ± 0.26	0.43
Traffic Load: Childhood, school	[0,6.33]	(6.33,96.7]	0.09 ± 0.27	0.75	(96.7,276]	0.14 ± 0.27	0.59
Traffic Density: Pregnancy	[0,7.94]	(7.94,13.9]	-0.42 ± 0.28	0.13	(13.9,39.3]	-0.21 ± 0.26	0.42
Traffic Density: Childhood, home	[0,10]	(10,19.8]	-0.04 ± 0.27	0.89	(19.8,49.3]	-0.35 ± 0.34	0.29
Traffic Density: Childhood, school	[0,10.5]	(10.5,19.6]	0.08 ± 0.27	0.78	(19.6,51.2]	0.16 ± 0.32	0.62
Indoor NO2: Childhood, home	[3.16,3.67]	(3.67,4.46]	0.24 ± 0.26	0.36	(4.46,6.96]	0.07 ± 0.36	0.85
Indoor PM2.5: Childhood, home	[2.53,2.78]	(2.78,2.98]	0.23 ± 0.25	0.36	(2.98,5.28]	0.29 ± 0.28	0.30
Indoor Benzene: Childhood, home	[0.737,1.06]	(1.06,1.26]	-0.21 ± 0.25	0.40	(1.26,3.16]	-0.32 ± 0.27	0.24
Indoor BTEX: Childhood, home	[3.03,3.41]	(3.41,3.71]	-0.01 ± 0.26	0.96	(3.71,5.04]	-0.07 ± 0.31	0.81

Note: BTEX, sum of benzene, toluene, ethylbenzene, and xylene; GGT, gamma-glutamyltransferase; NO2, nitrogen dioxide; PM2.5, particulate matter <2.5 µm; PM10, particulate matter <10 µm.

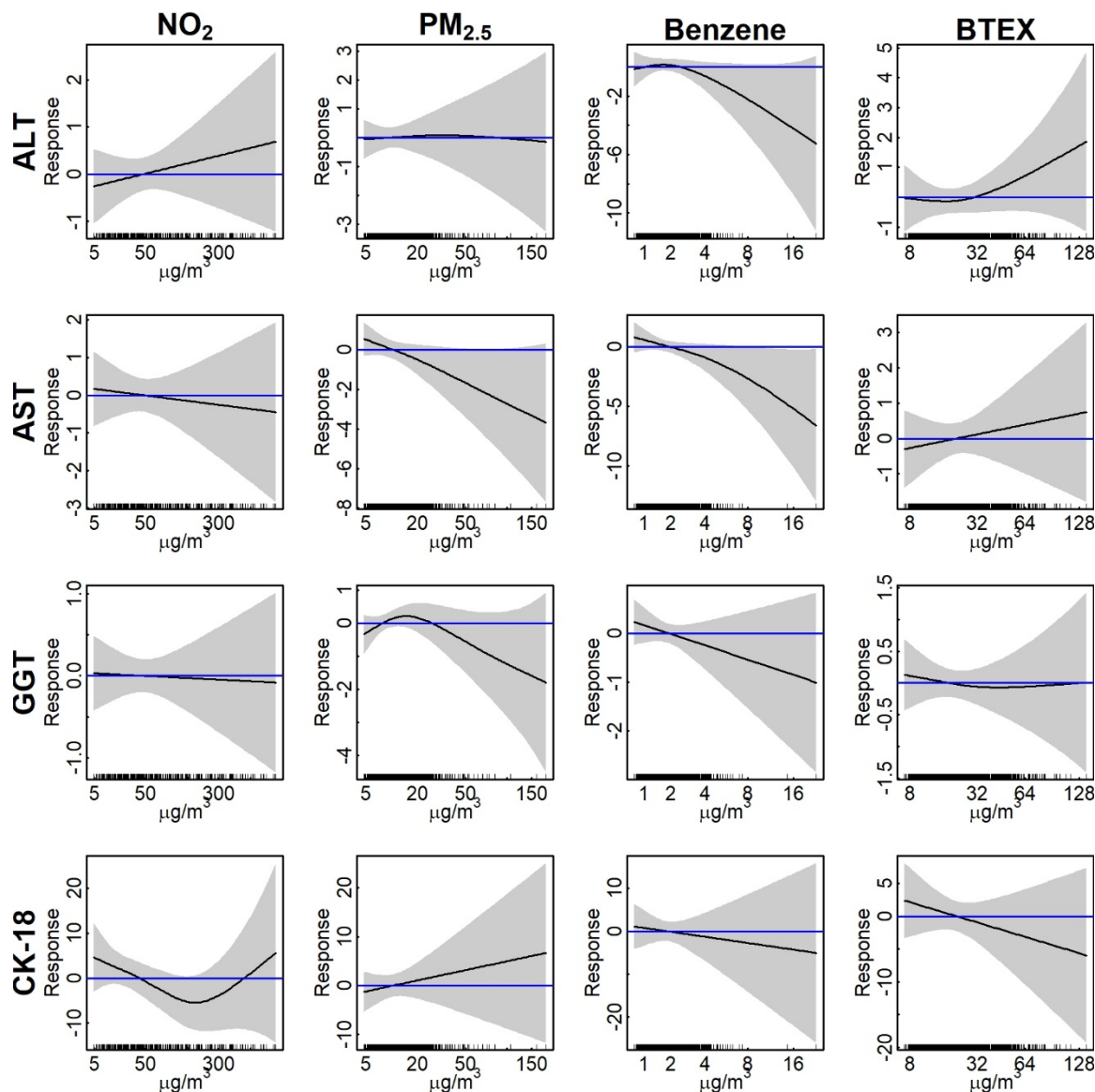
^a Models adjusted for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy.

^b Ranges shown are for transformed exposure values, when exposure was transformed.

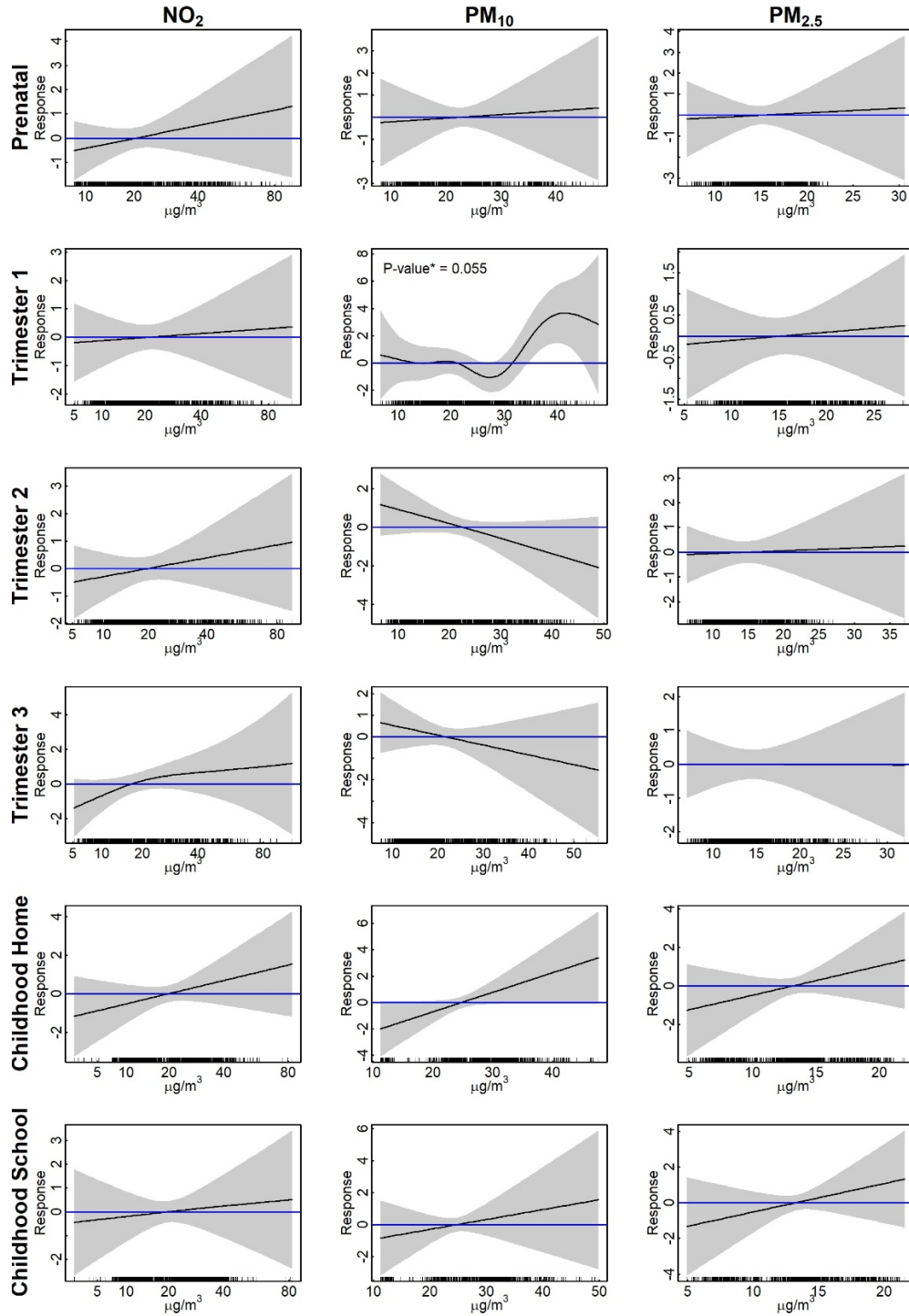
^c P values shown are unadjusted for multiple comparisons. No P values were statistically significant after adjustment for multiple comparisons.



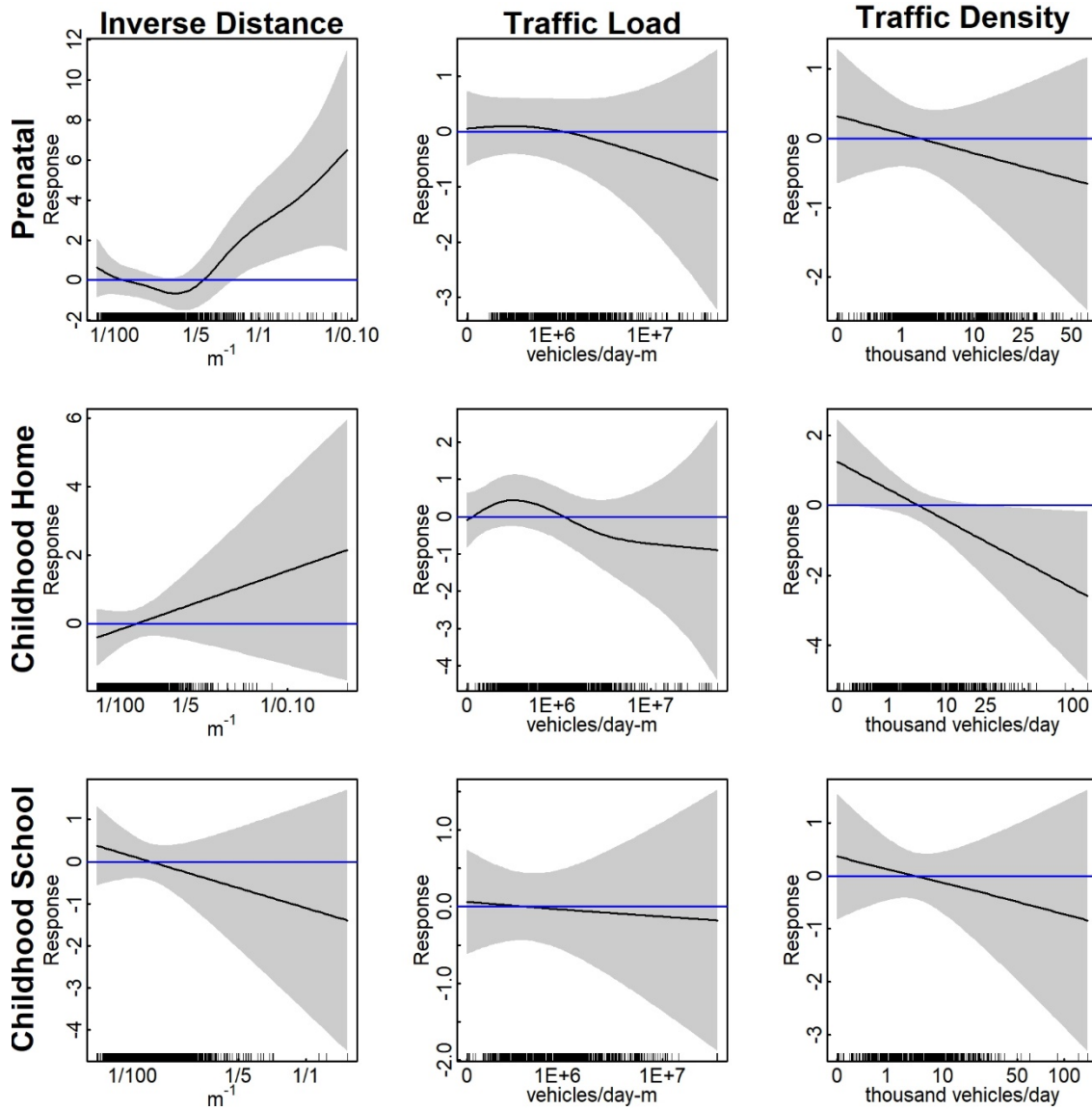
eFigure 1: Model results for traffic exposure and alanine aminotransferase (ALT). Associations between markers of traffic exposure during different age windows and ALT modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



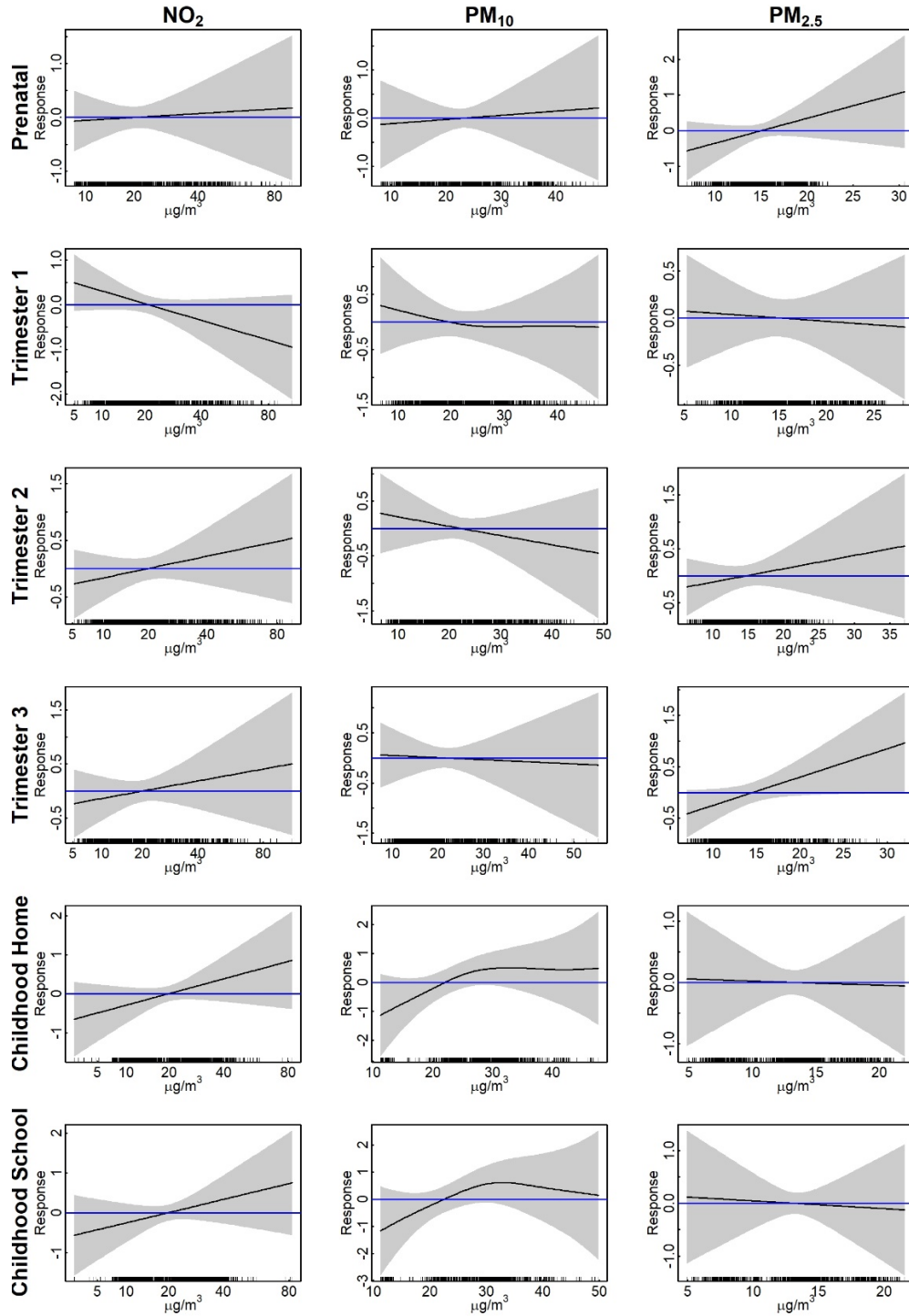
eFigure 2: Model results for indoor air pollution and liver enzyme biomarkers. Associations between indoor air pollution exposure (nitrogen dioxide [NO₂], particulate matter <2.5 µm [PM_{2.5}], benzene, and sum of benzene, toluene, ethylbenzene, and xylene [BTEX]) in childhood and liver enzyme biomarkers (alanine aminotransferase [ALT], aspartate aminotransferase [AST], gamma-glutamyltransferase [GGT], and cytokeratin-18 [CK-18]) modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



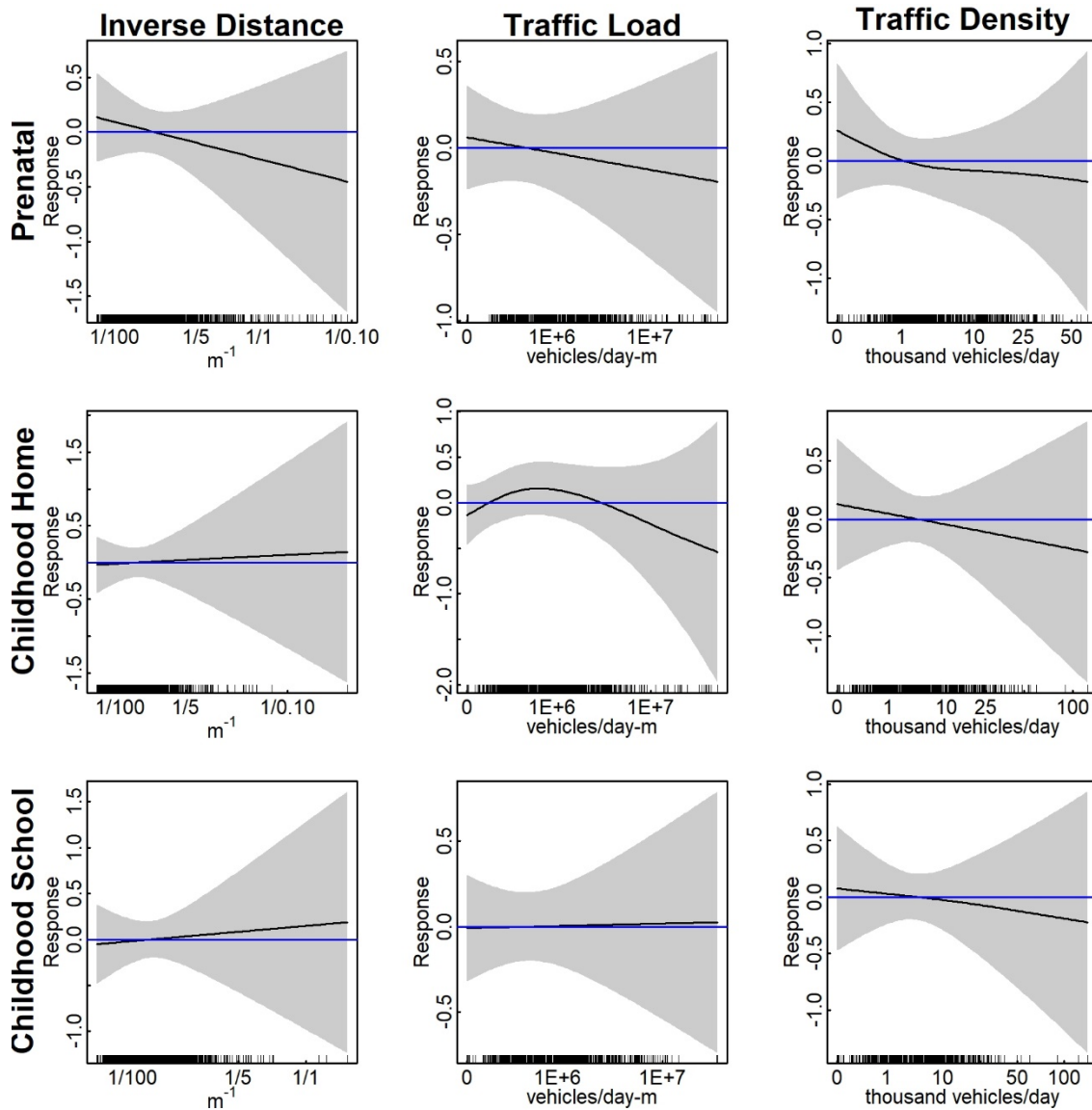
eFigure 3: Model results for ambient air pollution and aspartate aminotransferase (AST). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10 \mu\text{m}$ [PM_{10}], and particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and AST modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons. *Association for PM_{10} exposure in trimester 1 was marginally statistically significant after adjustment for multiple comparisons.



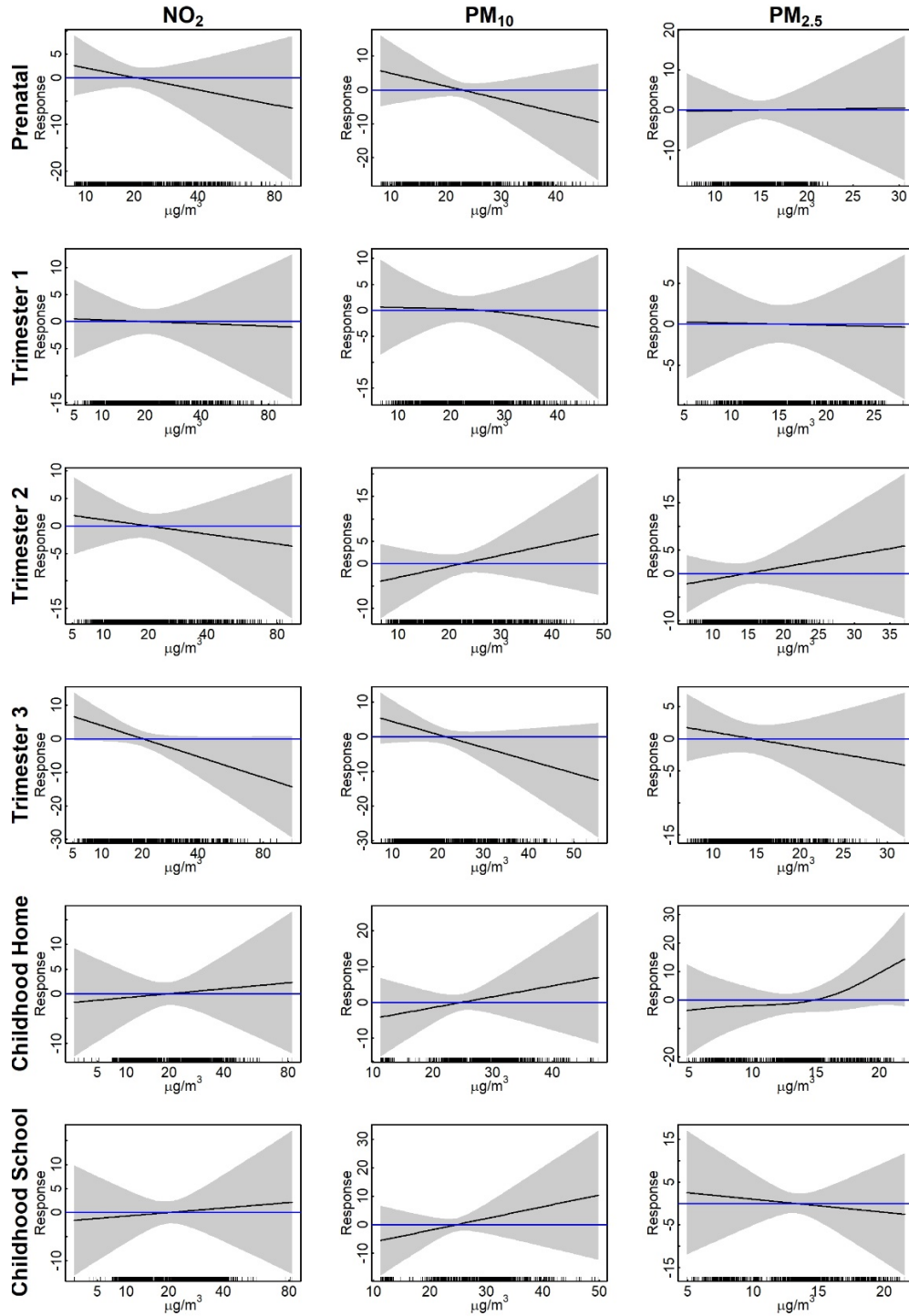
eFigure 4: Model results for traffic exposure and aspartate aminotransferase (AST). Associations between markers of traffic exposure during different age windows and AST modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



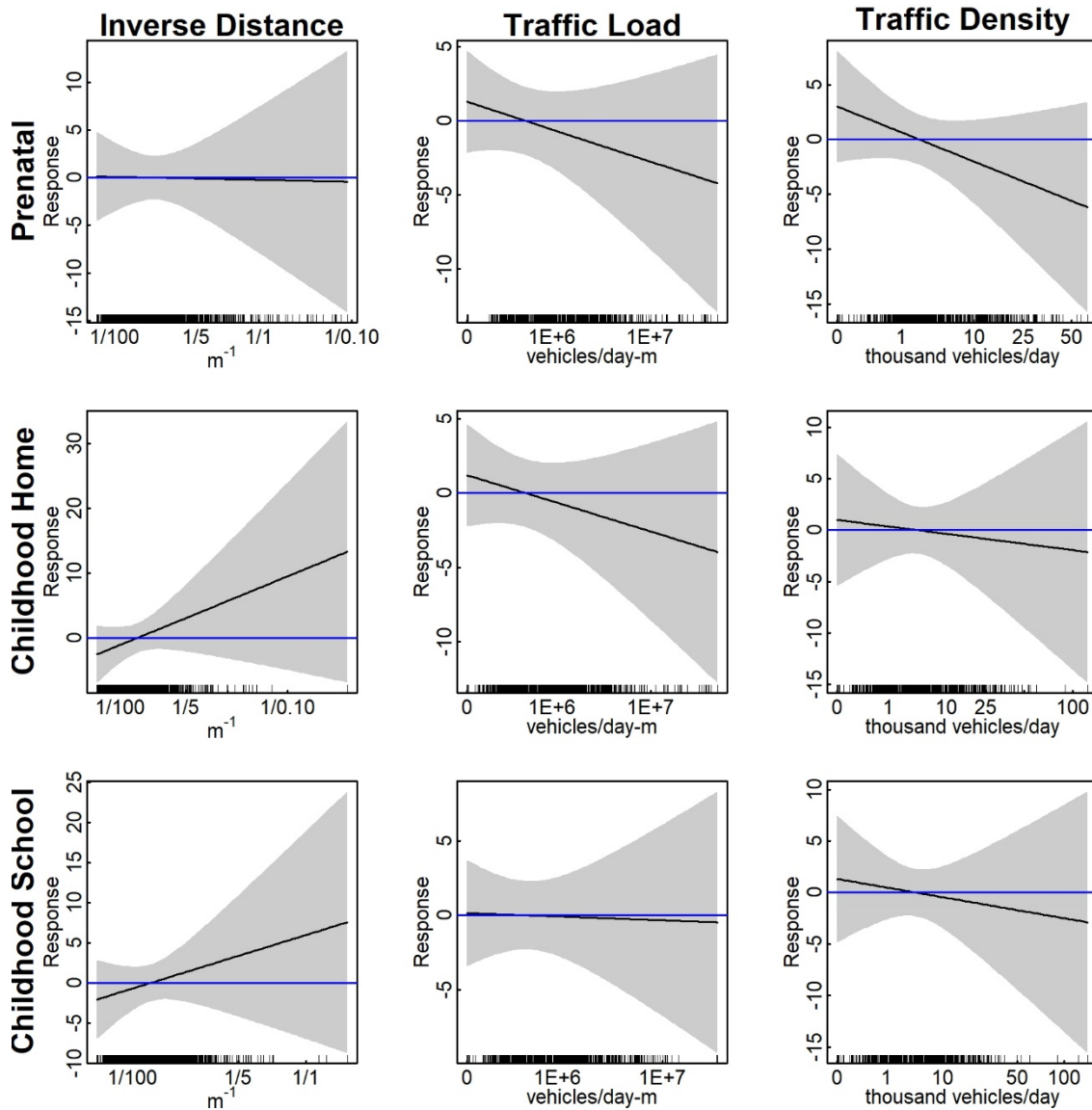
eFigure 5: Model results for ambient air pollution and gamma-glutamyltransferase (GGT). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10 \mu\text{m}$ [PM_{10}], and particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and GGT modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



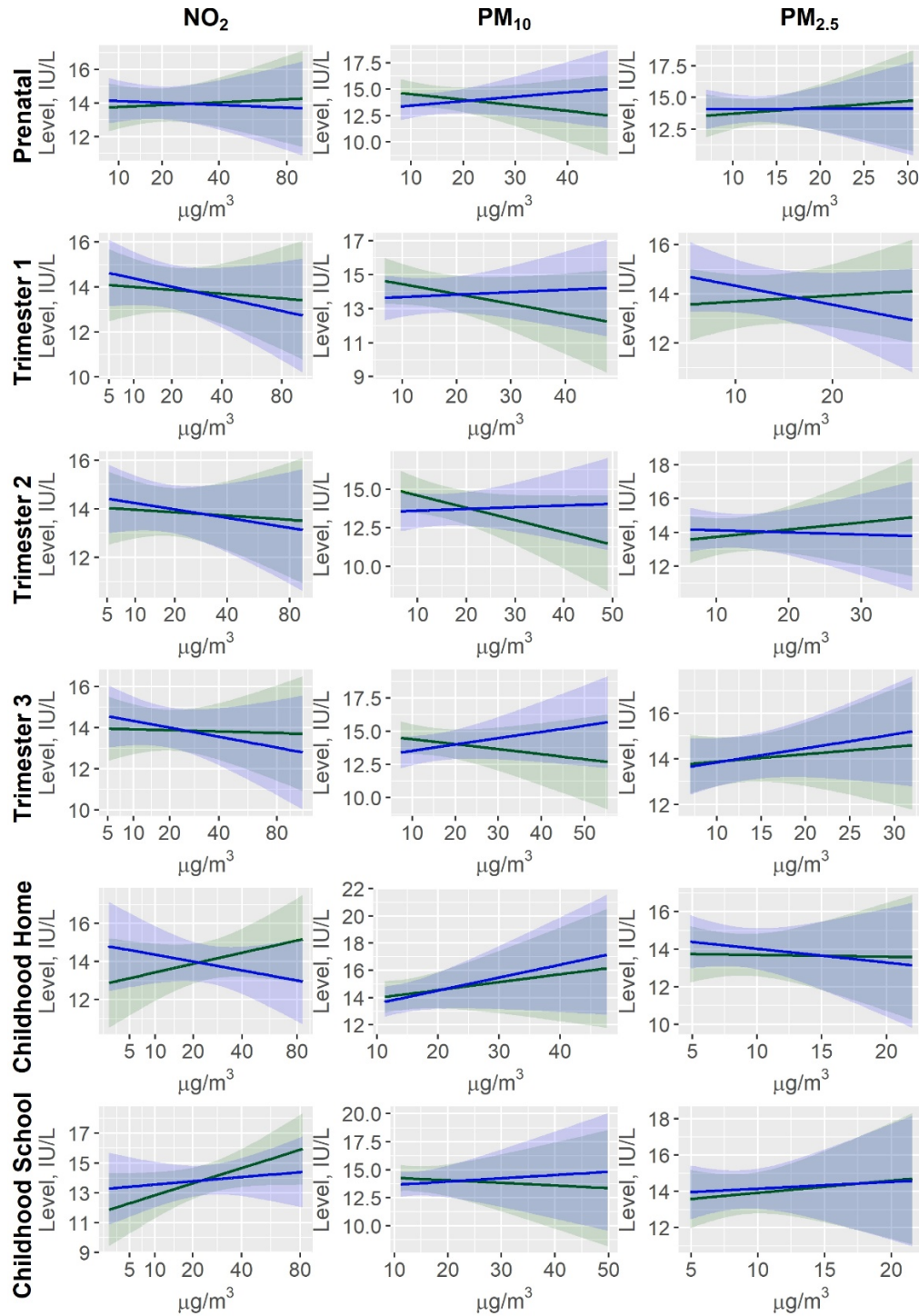
eFigure 6: Model results for traffic exposure and gamma-glutamyltransferase (GGT). Associations between markers of traffic exposure during different age windows and GGT modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



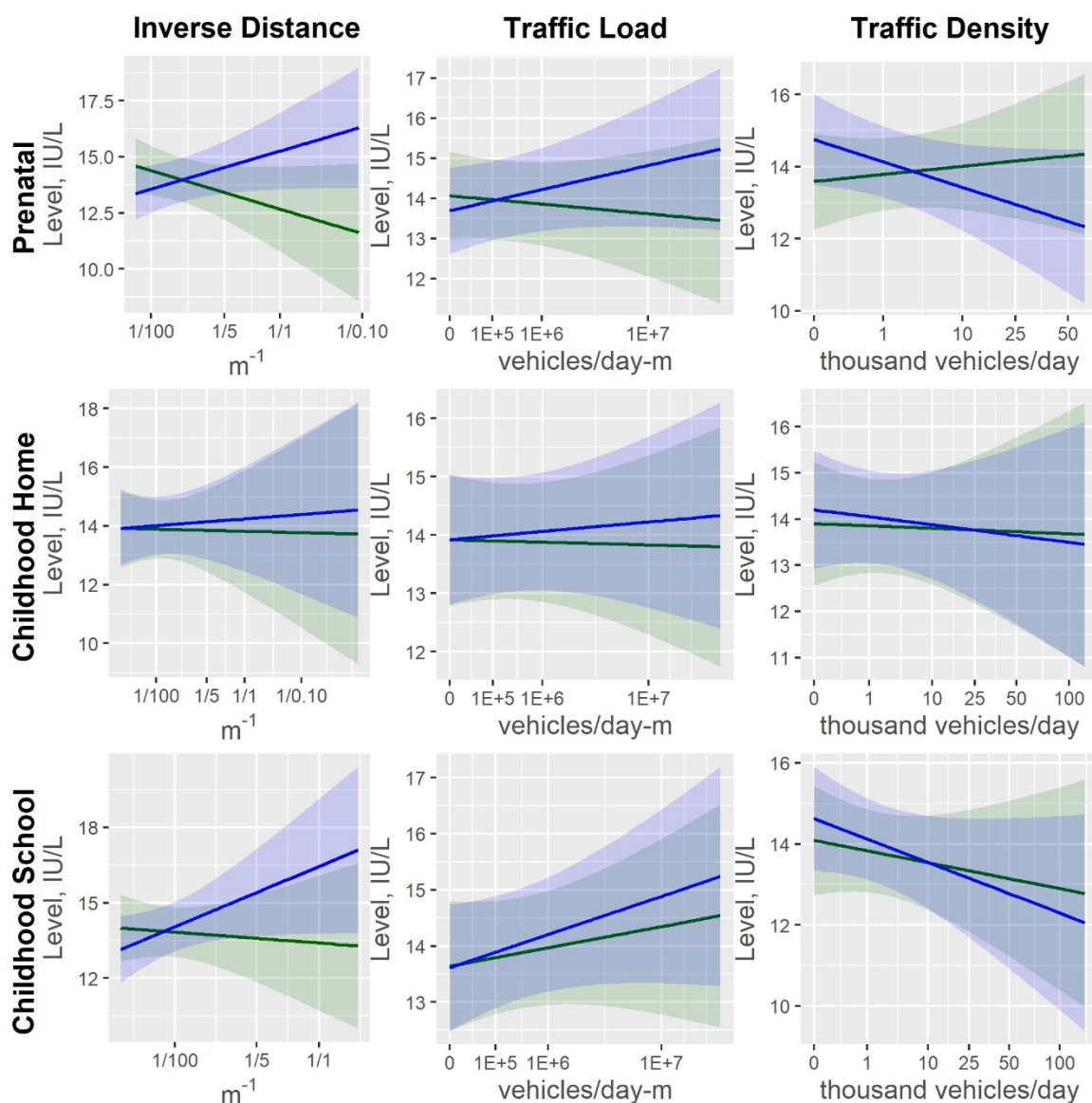
eFigure 7: Model results for ambient air pollution and cytochrome P450 1A2 (CYP1A2). Associations between ambient air pollution exposure (nitrogen dioxide [NO₂], particulate matter <10 µm [PM₁₀], and particulate matter <2.5 µm [PM_{2.5}]) during different age windows and CYP1A2 modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null reference line shown. No results were statistically significant after adjustment for multiple comparisons.



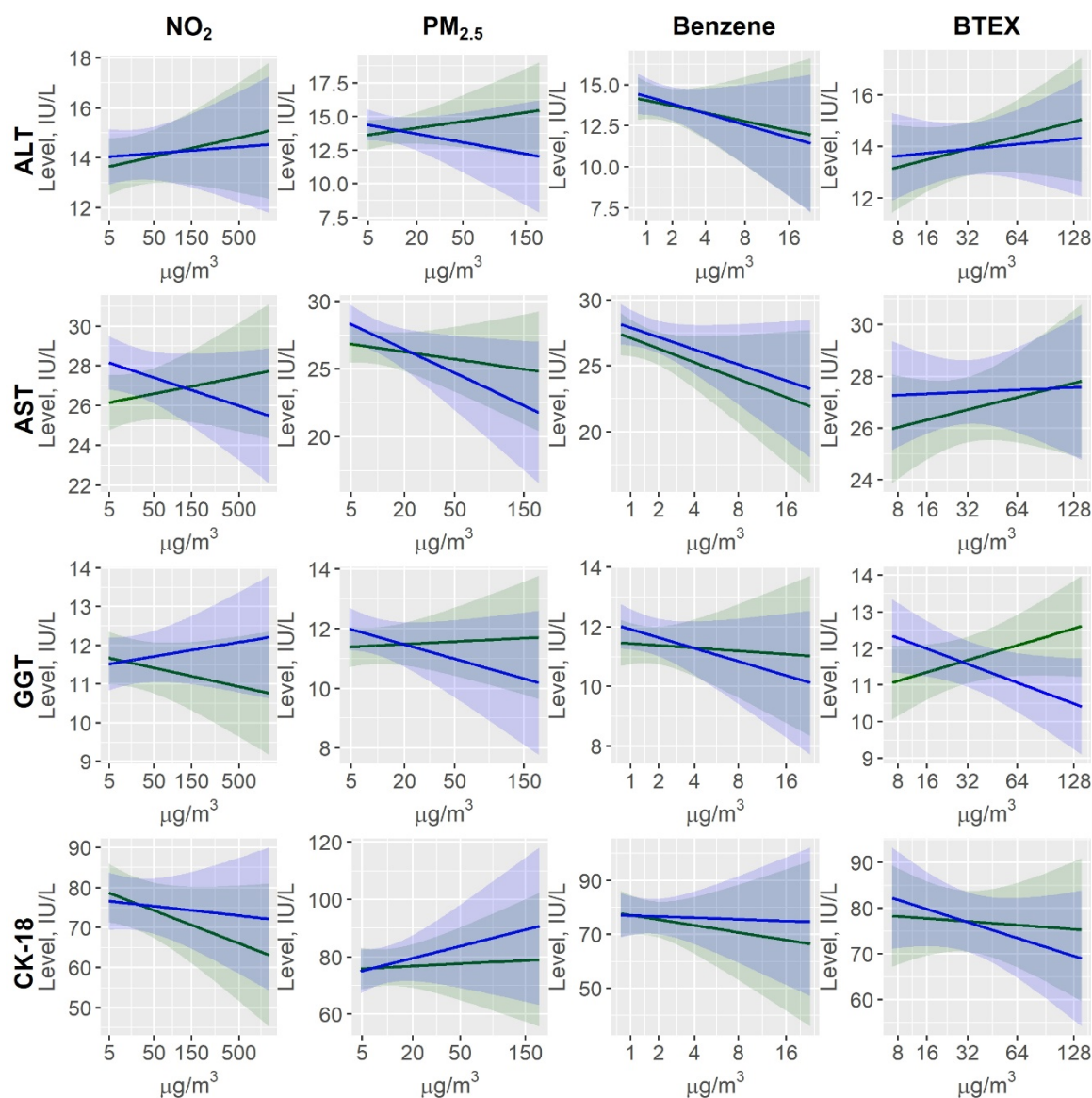
eFigure 8: Model results for traffic exposure and cytokeratin-18 (CK-18). Associations between markers of traffic exposure during different age windows and CK-18 modeled separately using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. Null referent line shown. No results were statistically significant after adjustment for multiple comparisons.



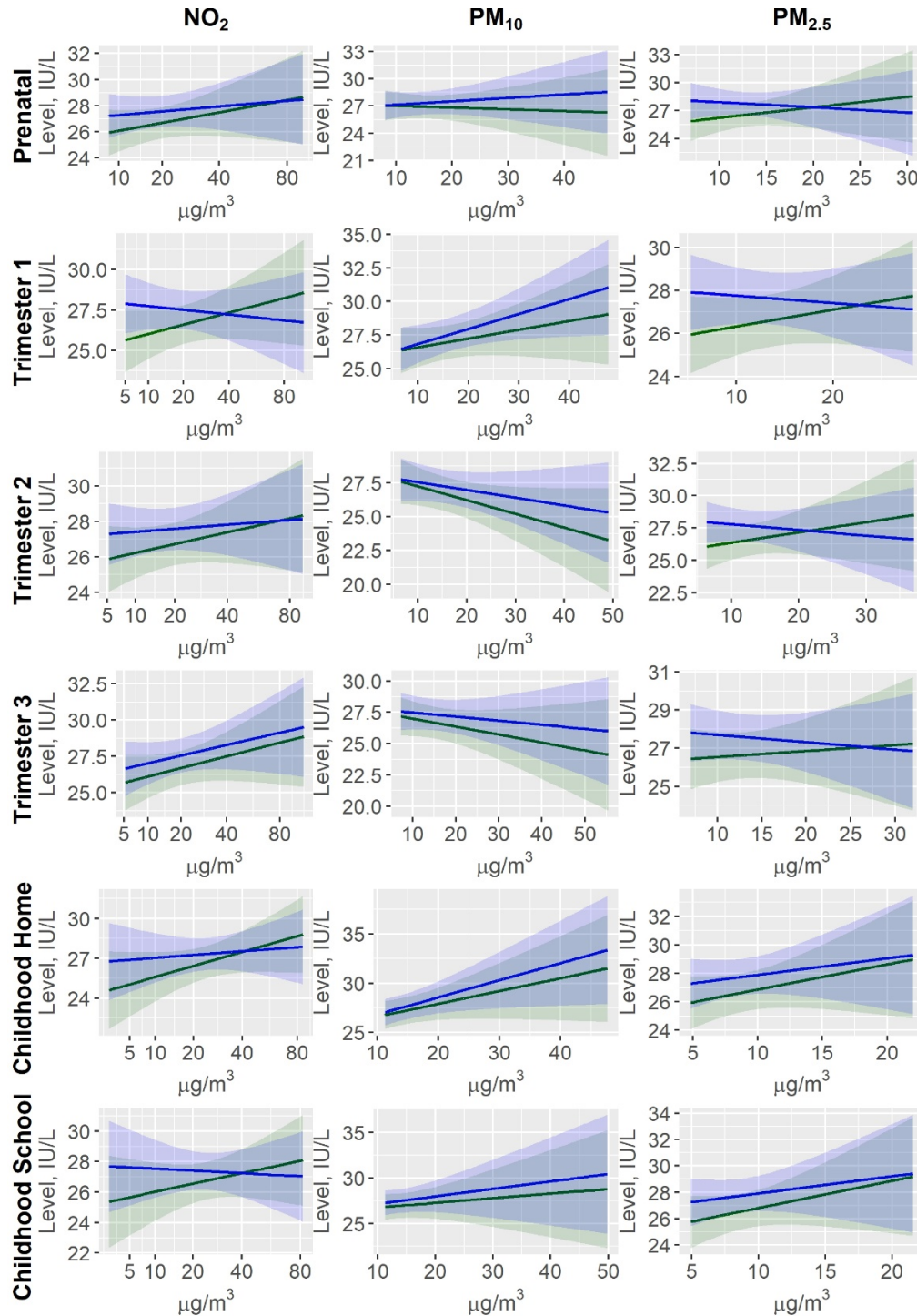
eFigure 9: Model results for ambient air pollution with sex interaction and alanine aminotransferase (ALT). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10 \mu\text{m}$ [PM_{10}], and particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and ALT, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



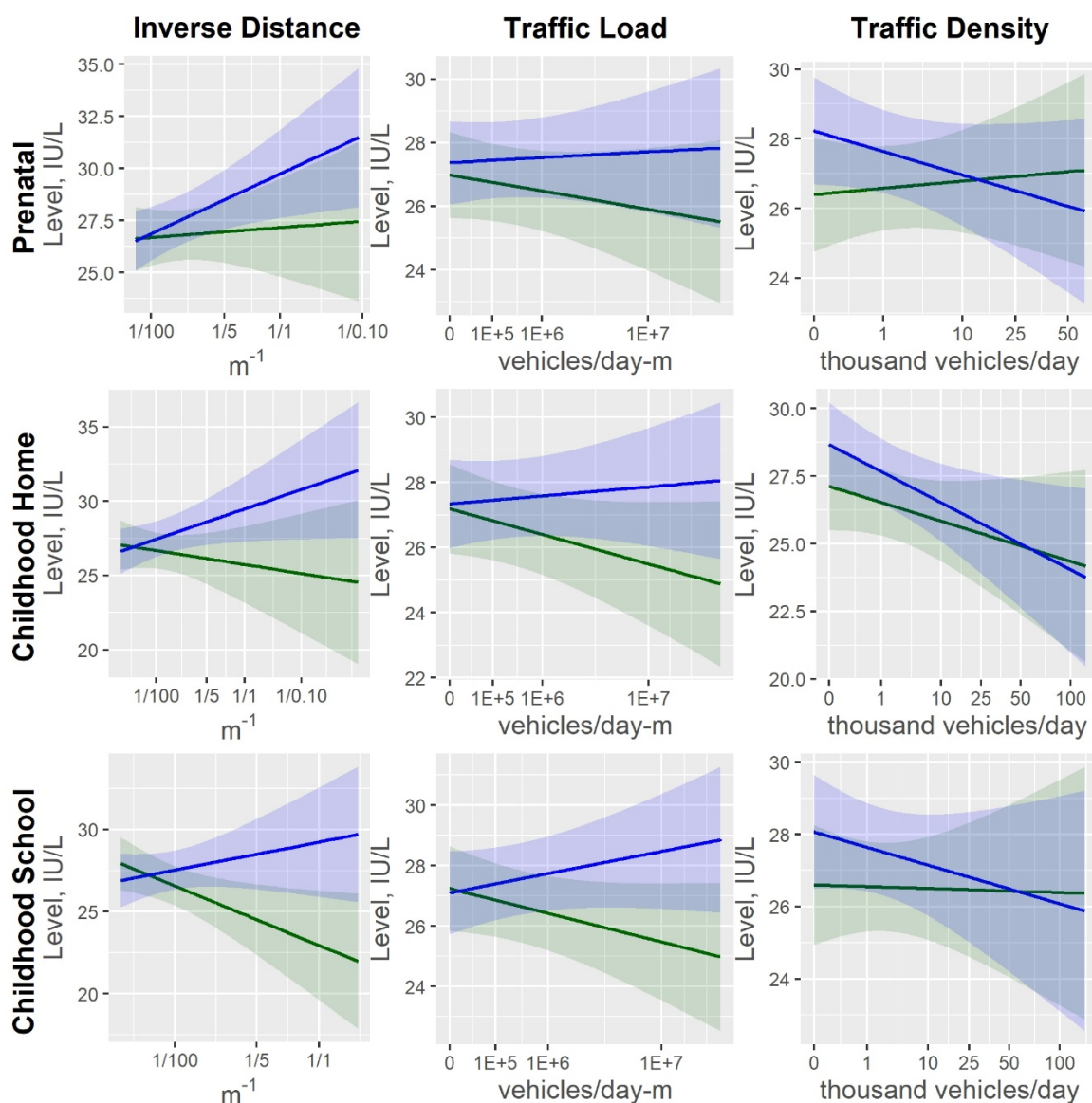
eFigure 10: Model results for traffic exposure with sex interaction and alanine aminotransferase (ALT). Associations between markers of traffic exposure during different age windows and ALT, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



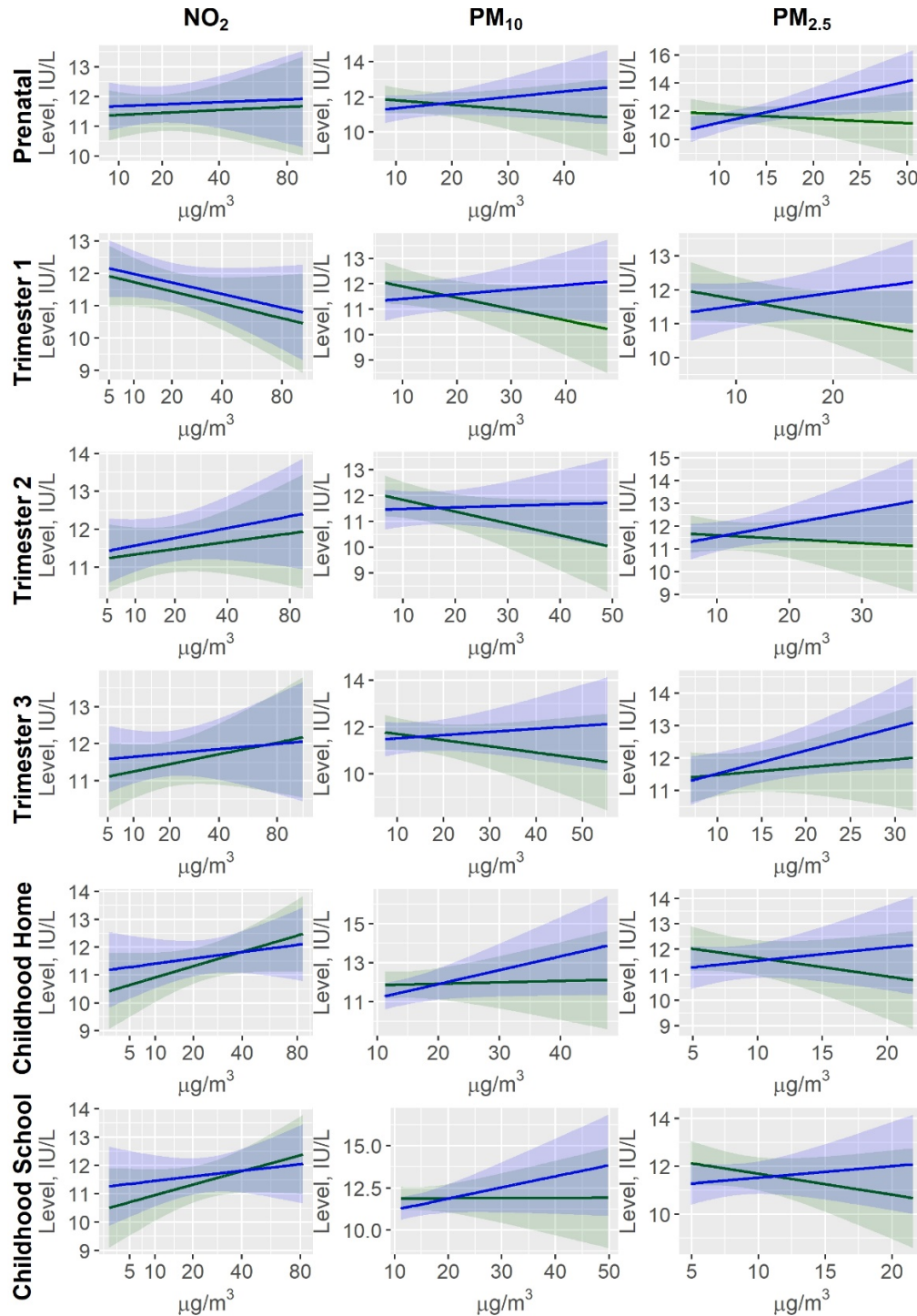
eFigure 11: Model results for indoor air pollution with sex interaction and liver enzyme biomarkers. Associations between indoor air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$], benzene, and sum of benzene, toluene, ethylbenzene, and xylene [BTEX]) in childhood and liver enzyme biomarkers (alanine aminotransferase [ALT], aspartate aminotransferase [AST], gamma-glutamyltransferase [GGT], and cytokeratin-18 [CK-18]), stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



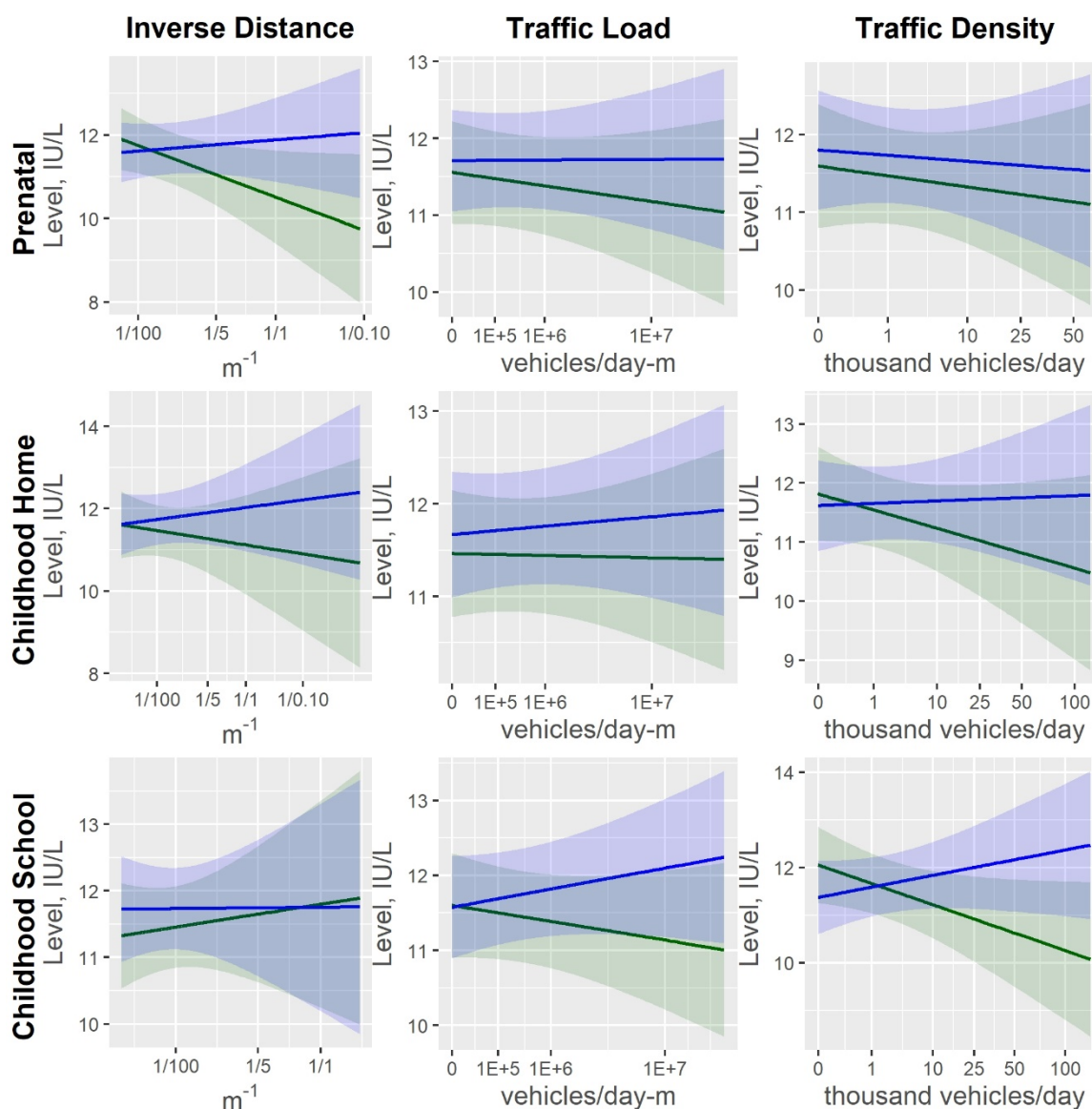
eFigure 12: Model results for ambient air pollution with sex interaction and aspartate aminotransferase (AST). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10 \mu\text{m}$ [PM_{10}], and particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and AST, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



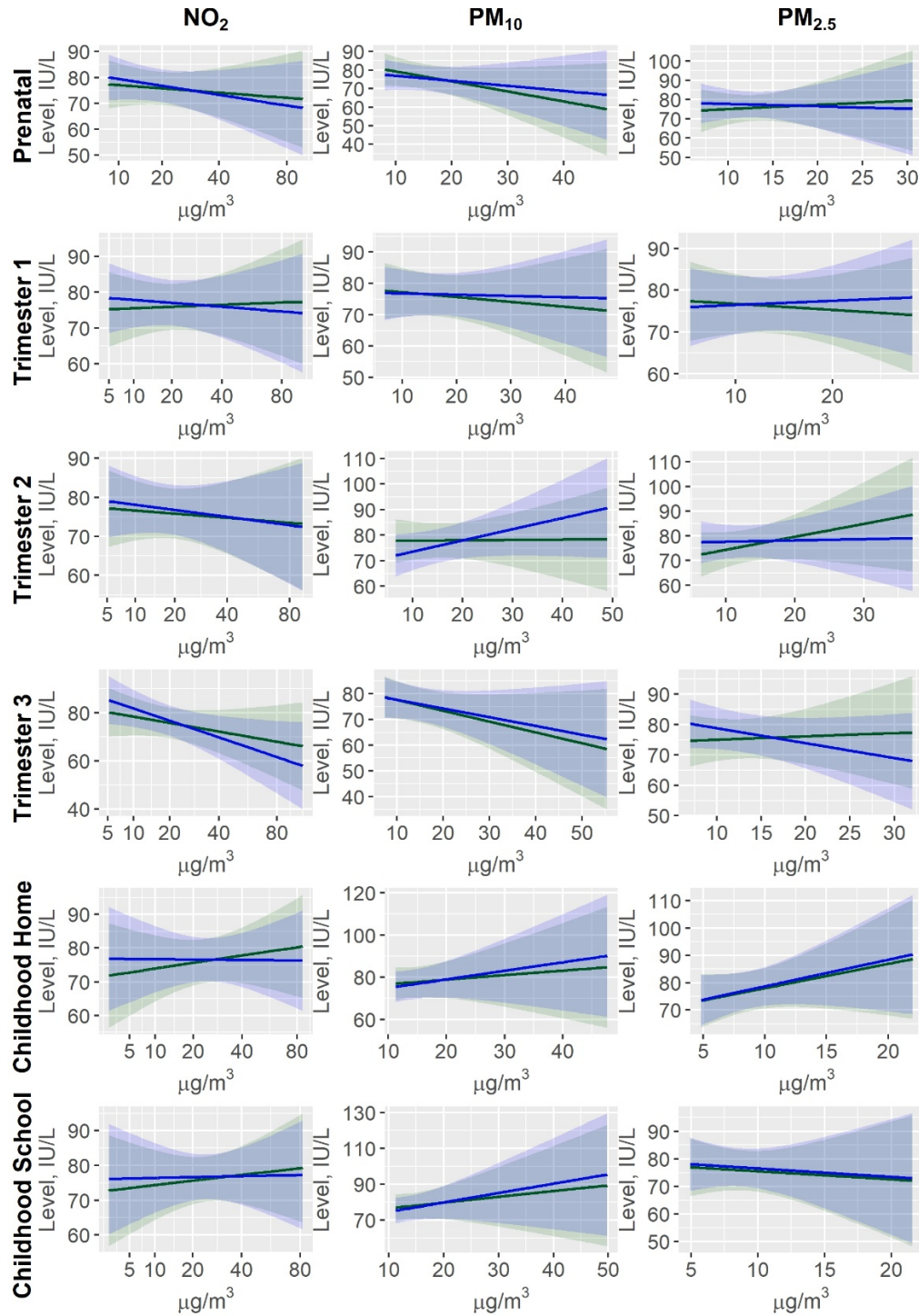
eFigure 13: Model results for traffic exposure with sex interaction and aspartate aminotransferase (AST). Associations between markers of traffic exposure during different age windows and AST, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



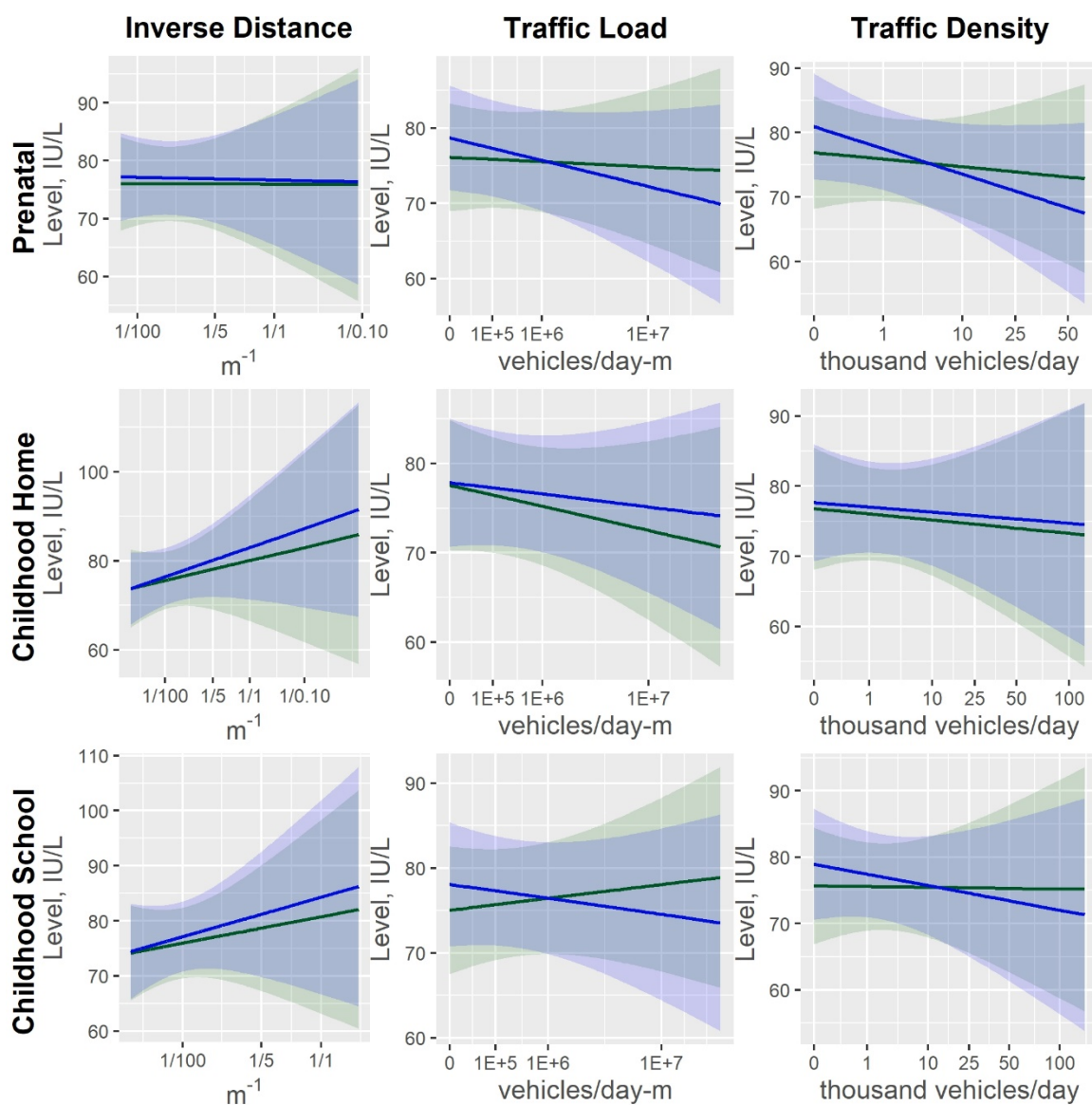
eFigure 14: Model results for ambient air pollution with sex interaction and gamma-glutamyltransferase (GGT). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10\ \mu\text{m}$ [PM_{10}], and particulate matter $<2.5\ \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and GGT, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



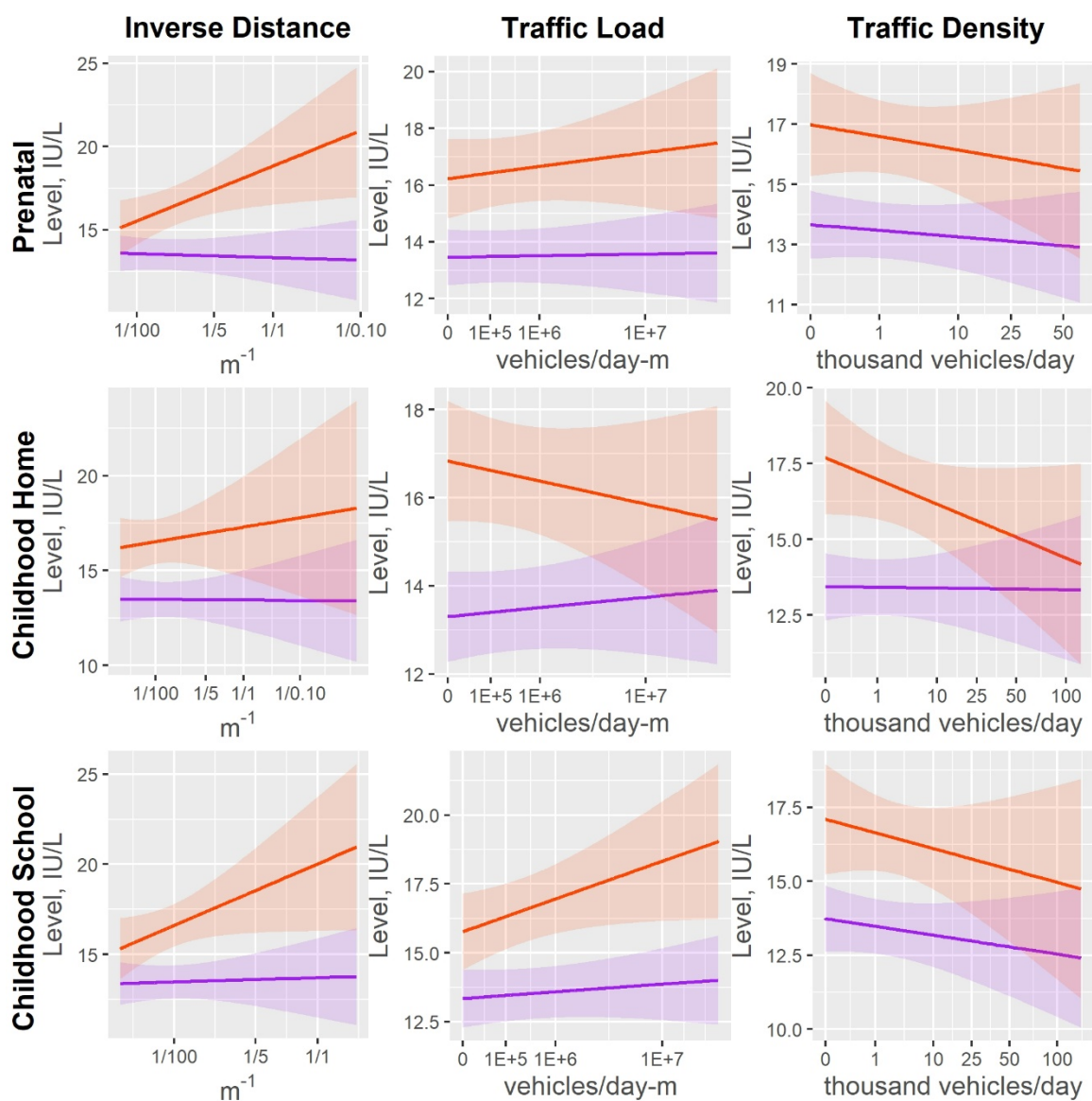
eFigure 15: Model results for traffic exposure with sex interaction and gamma-glutamyltransferase (GGT). Associations between markers of traffic exposure during different age windows and GGT, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



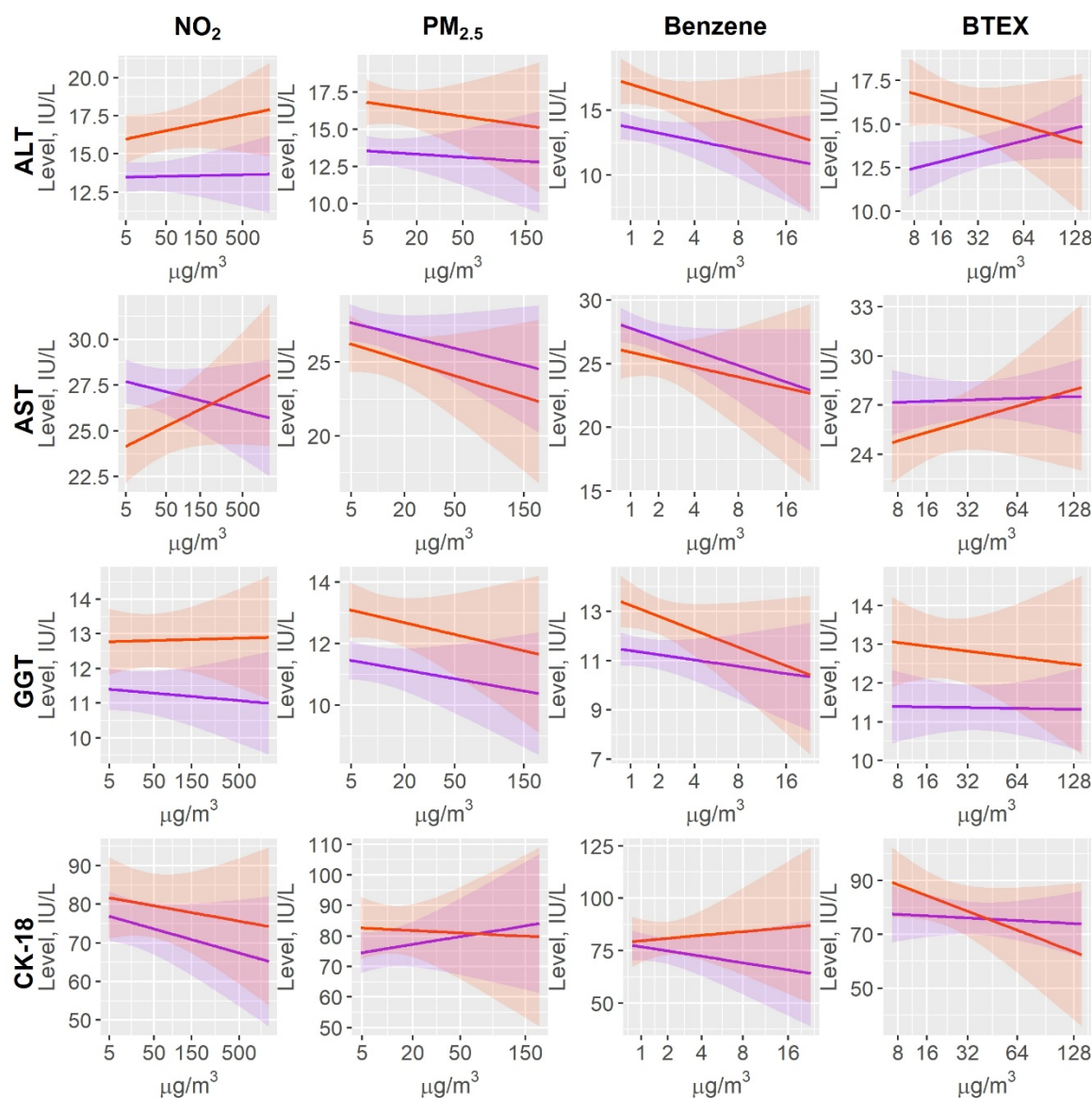
eFigure 16: Model results for ambient air pollution with sex interaction and cytokerin-18 (CK-18). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10\ \mu\text{m}$ [PM_{10}], and particulate matter $<2.5\ \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and CK-18, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



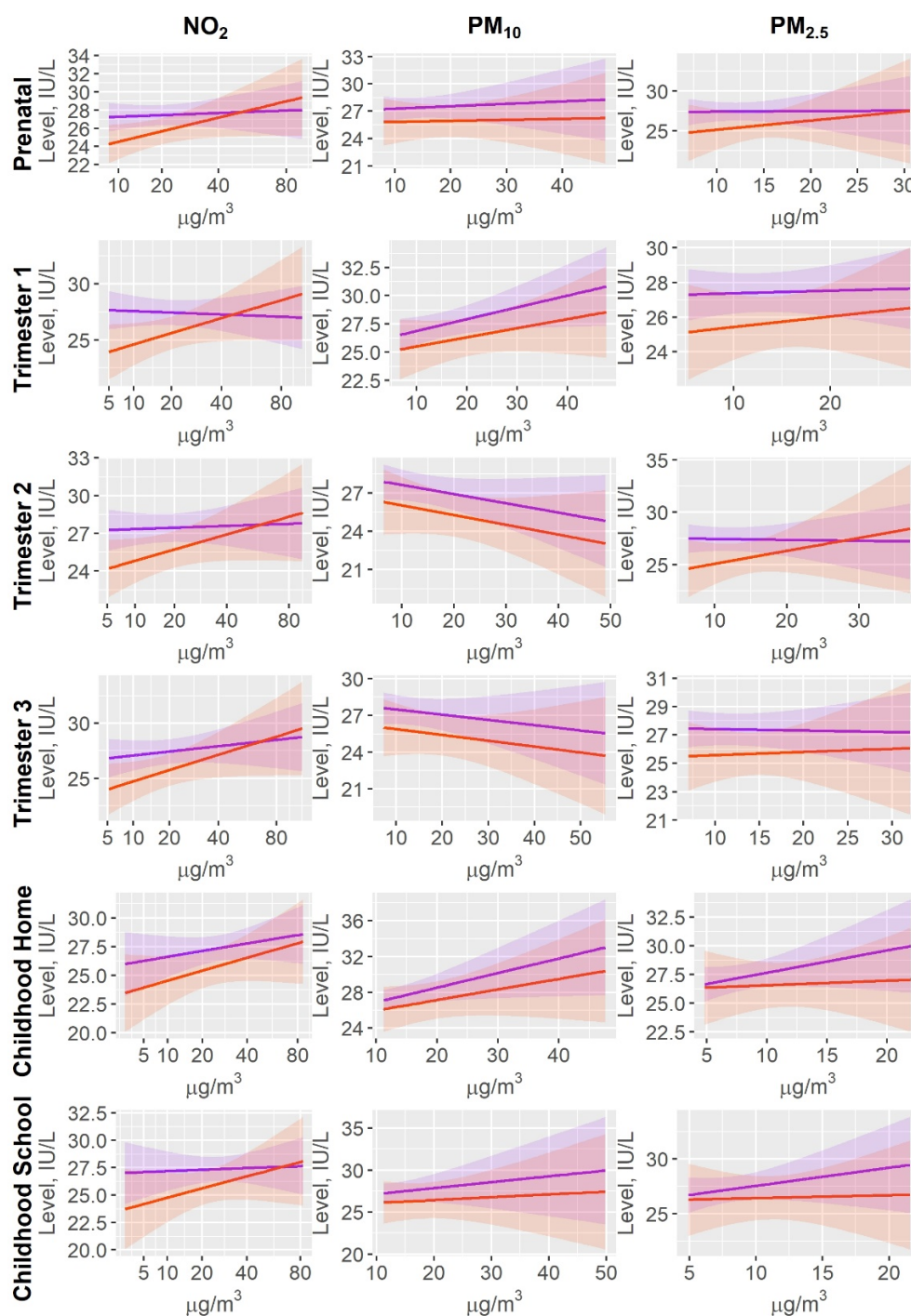
eFigure 17: Model results for traffic exposure with sex interaction and cytokeratin-18 (CK-18). Associations between markers of traffic exposure during different age windows and CK-18, stratified by sex (green line represents females; blue line represents males). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



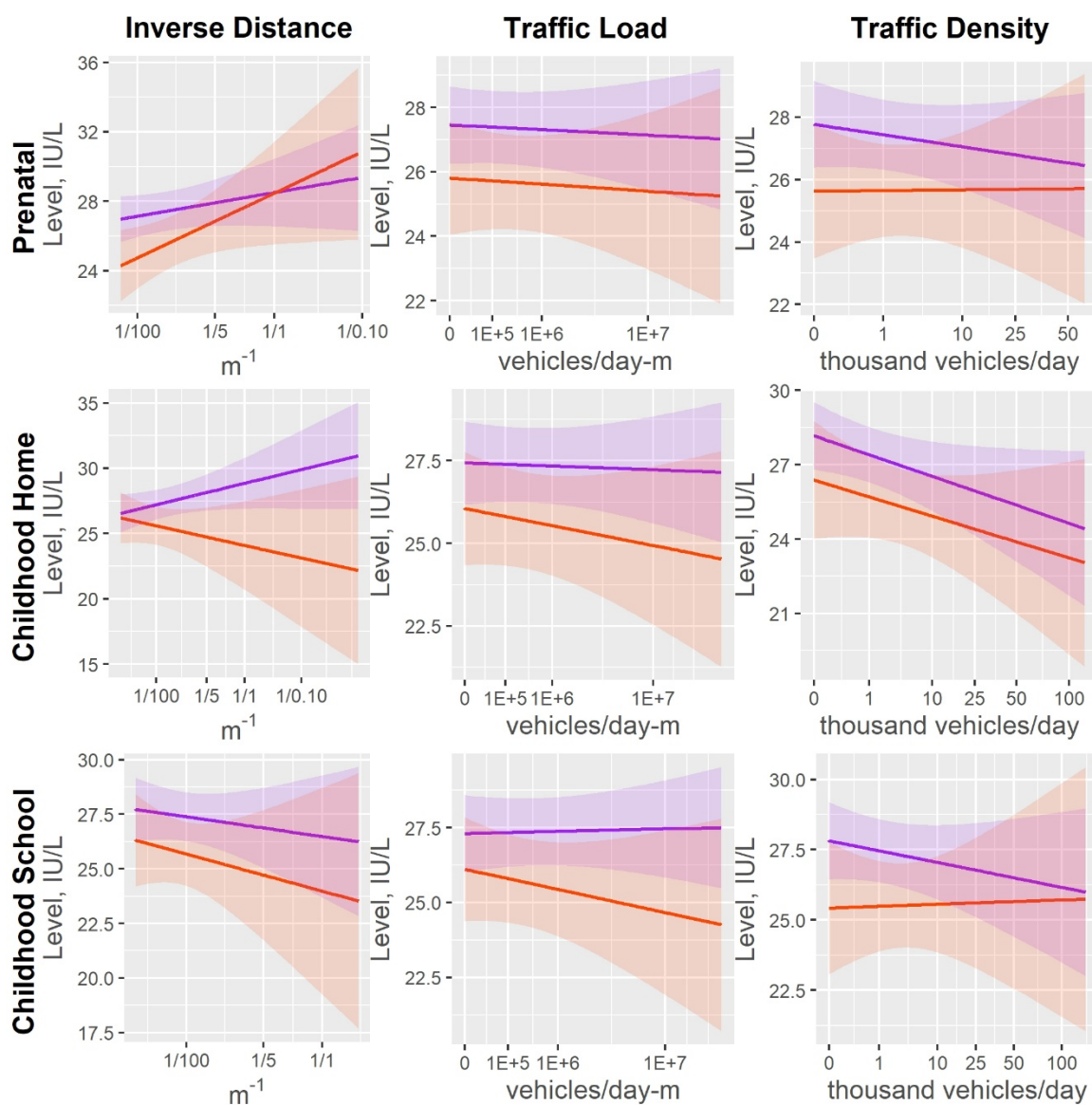
eFigure 18: Model results for traffic exposure with overweight or obese status interaction and alanine aminotransferase (ALT). Associations between markers of traffic exposure during different age windows and ALT, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



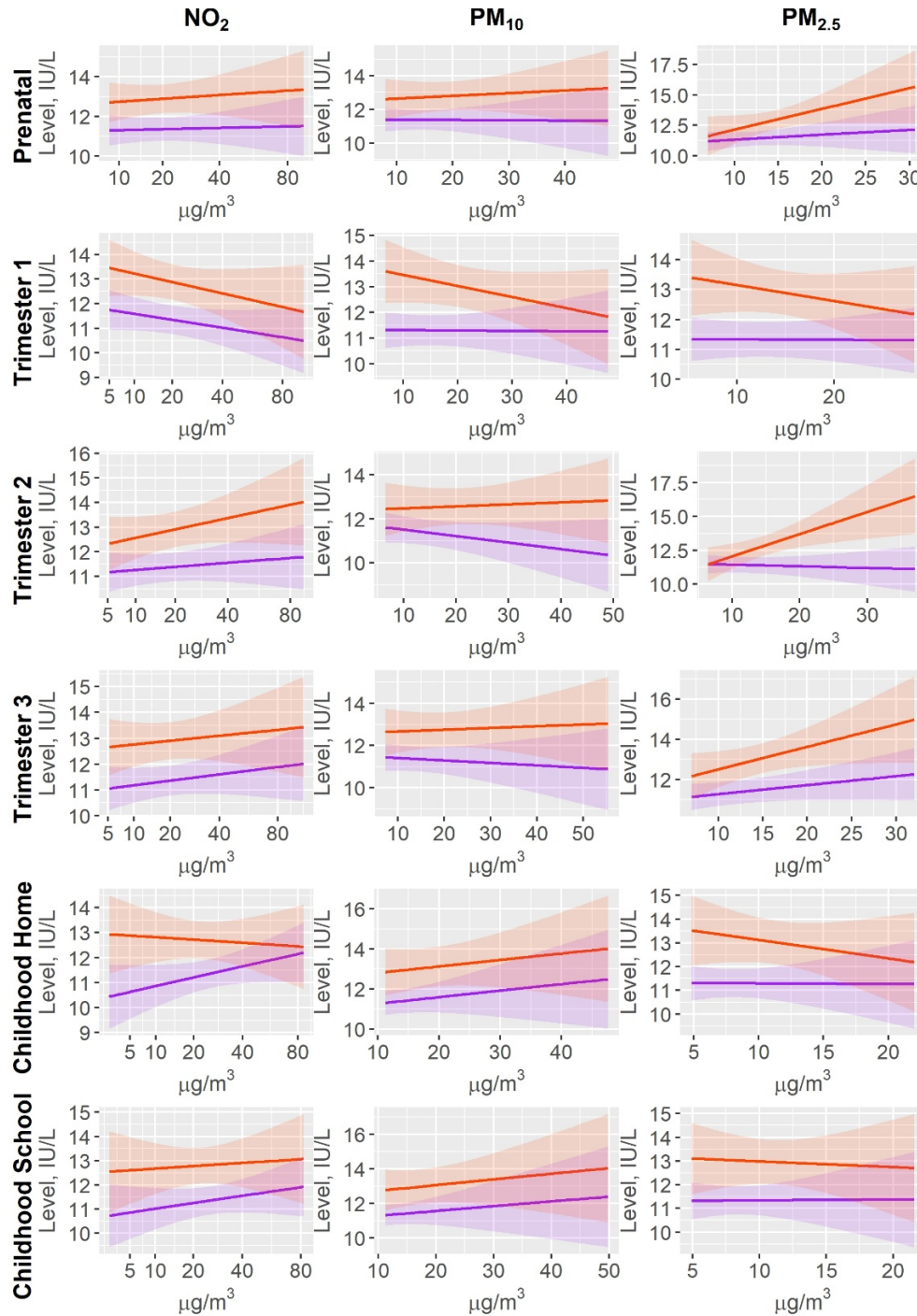
eFigure 19: Model results for indoor air pollution with overweight or obese status interaction and liver enzyme biomarkers. Associations indoor air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<2.5 \mu\text{m}$ [$\text{PM}_{2.5}$], benzene, and sum of benzene, toluene, ethylbenzene, and xylene [BTEX]) in childhood and between liver enzyme biomarkers (alanine aminotransferase [ALT], aspartate aminotransferase [AST], gamma-glutamyltransferase [GGT], and cytokeratin-18 [CK-18]), stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



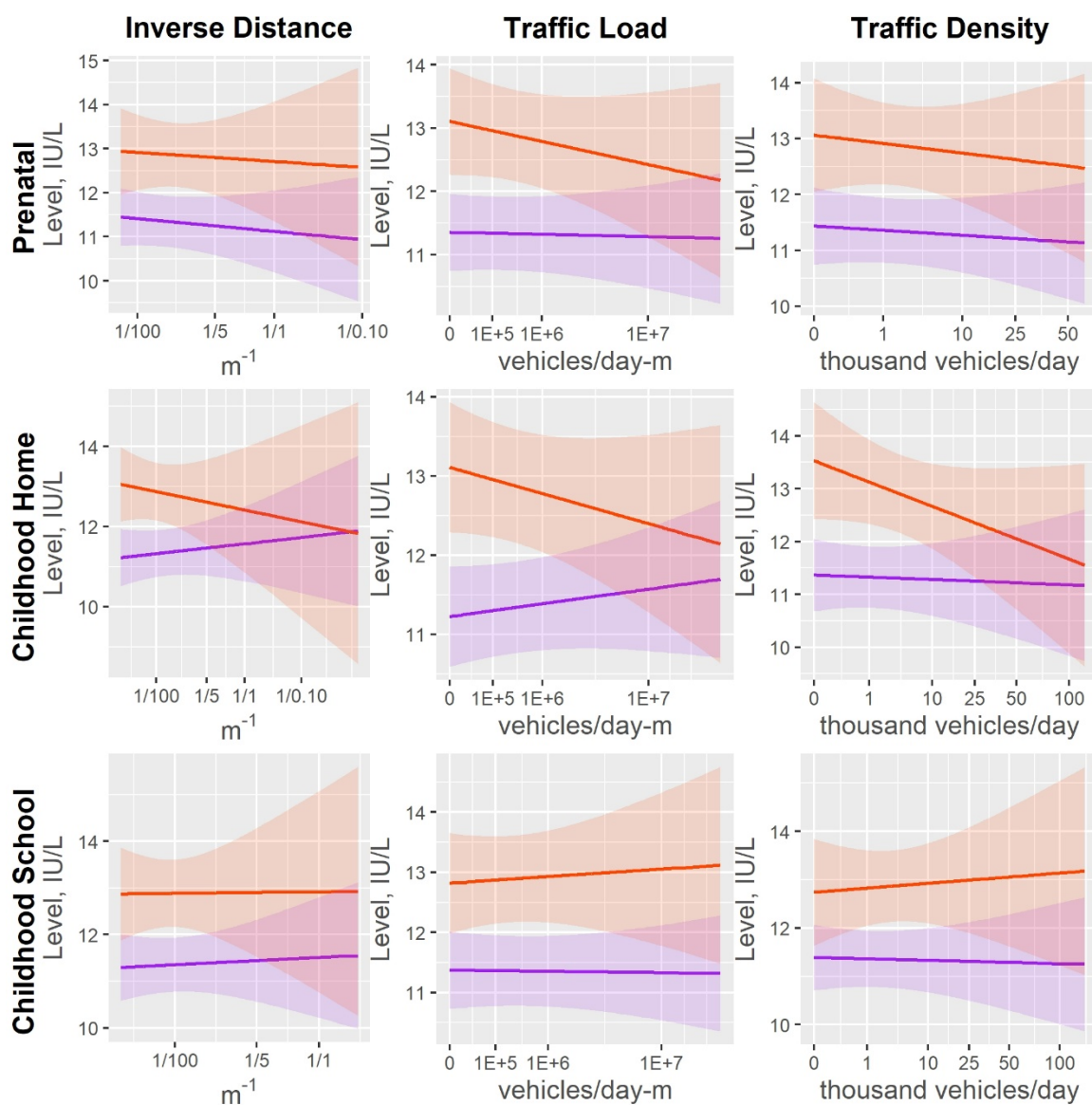
eFigure 20: Model results for ambient air pollution with overweight or obese status interaction and aspartate aminotransferase (AST). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10\ \mu\text{m}$ [PM_{10}], and particulate matter $<2.5\ \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and AST, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



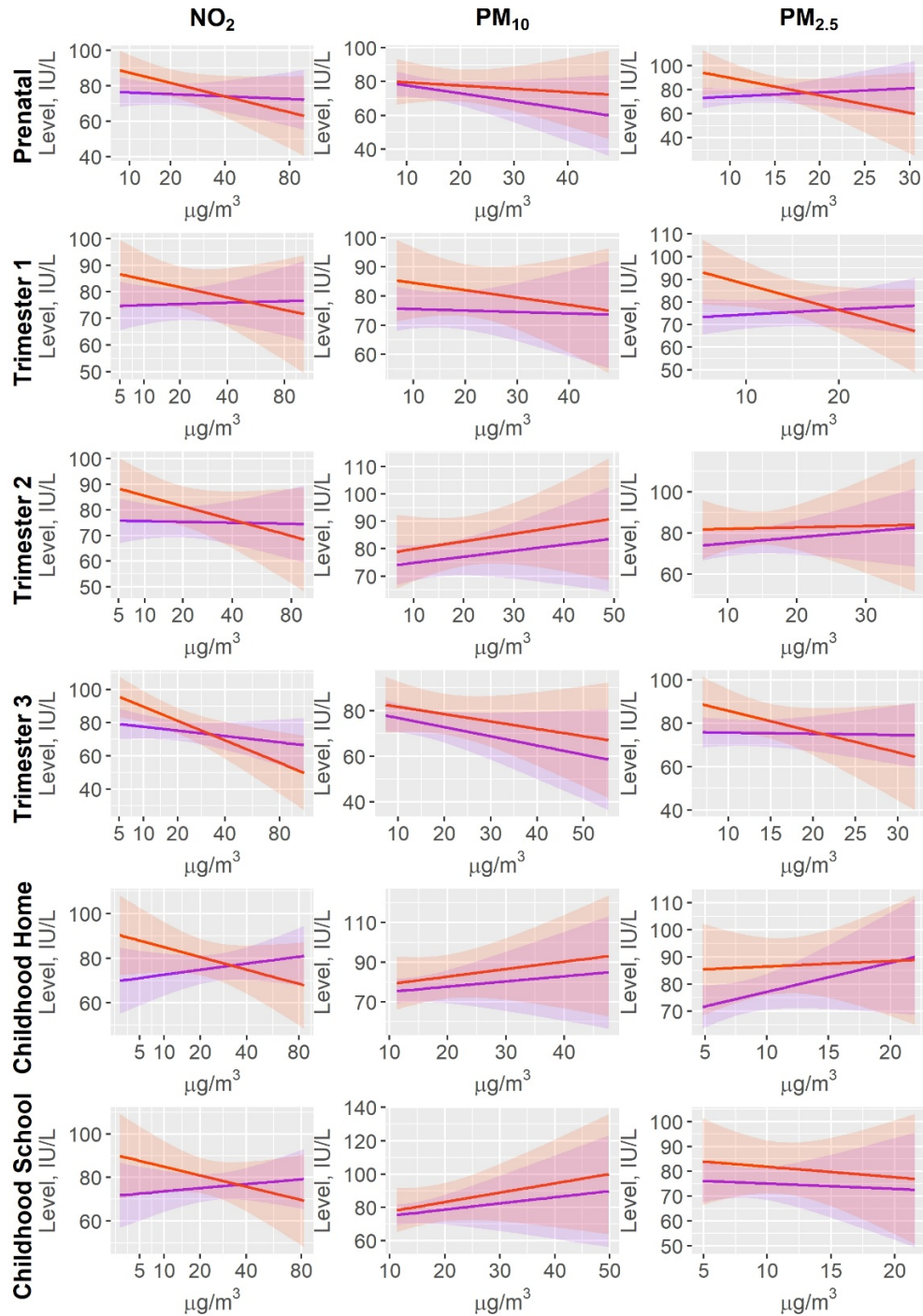
eFigure 21: Model results for traffic exposure with overweight or obese status interaction and aspartate aminotransferase (AST). Associations between markers of traffic exposure during different age windows and AST, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



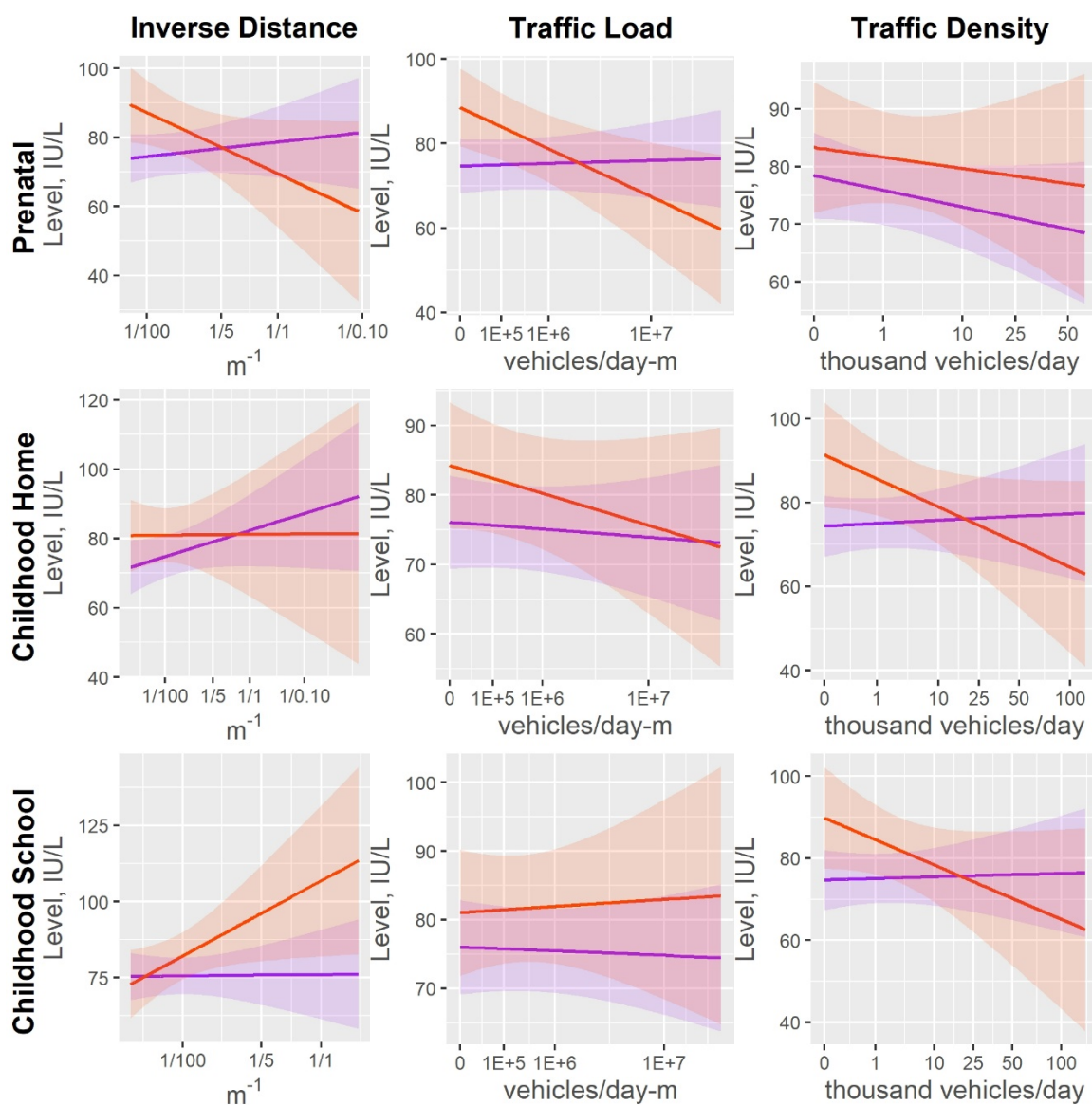
eFigure 22: Model results for ambient air pollution with overweight or obese status interaction and gamma-glutamyltransferase (GGT). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10\ \mu\text{m}$ [PM_{10}], and particulate matter $<2.5\ \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and GGT, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



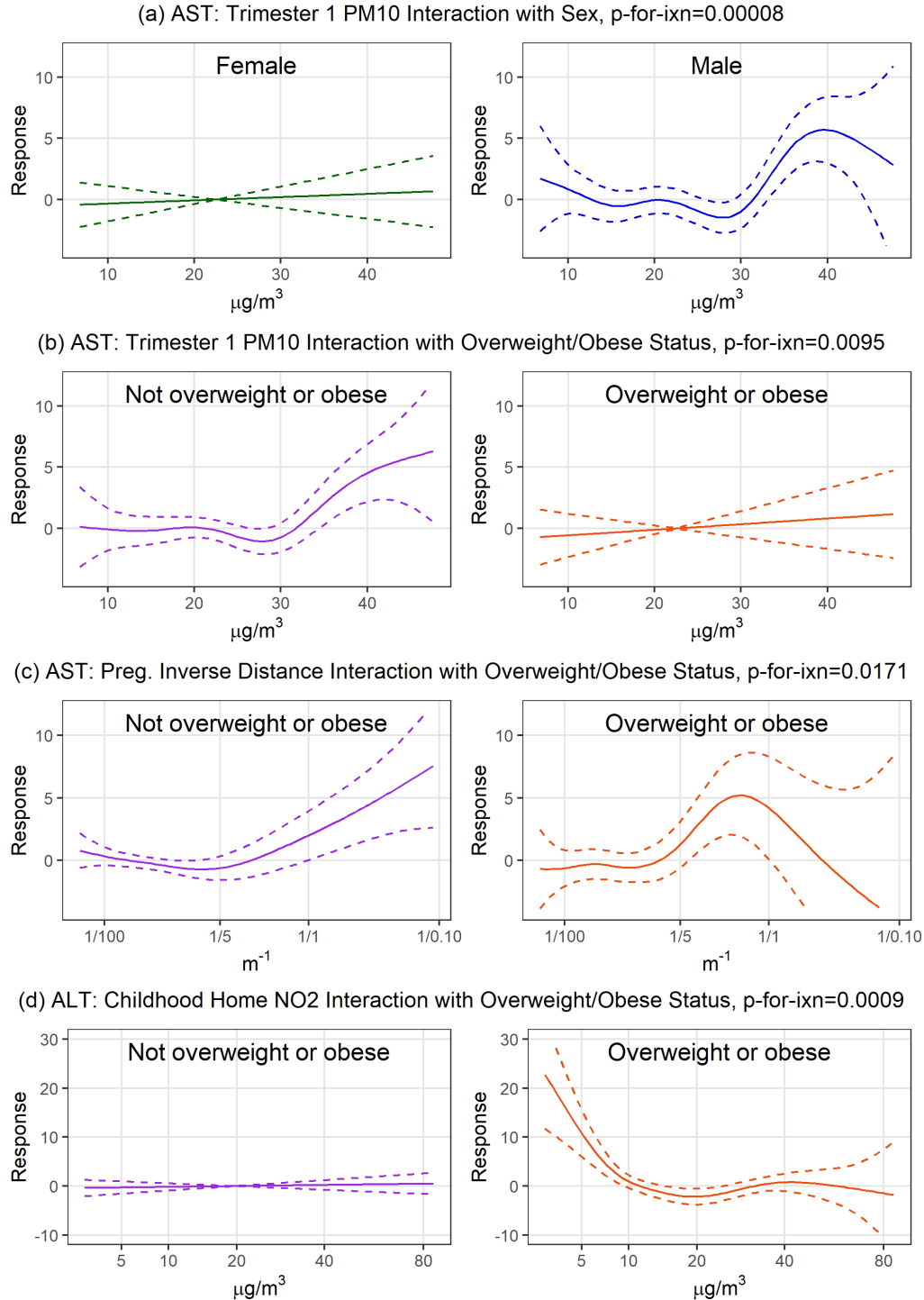
eFigure 23: Model results for traffic exposure with overweight or obese status interaction and gamma-glutamyltransferase (GGT). Associations between markers of traffic exposure during different age windows and GGT, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



eFigure 24: Model results for ambient air pollution with overweight or obese status interaction and cytokerin-18 (CK-18). Associations between ambient air pollution exposure (nitrogen dioxide [NO_2], particulate matter $<10\ \mu\text{m}$ [PM_{10}], and particulate matter $<2.5\ \mu\text{m}$ [$\text{PM}_{2.5}$]) during different age windows and CK-18, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



eFigure 25: Model results for traffic exposure with overweight or obese status interaction and cytokeratin-18 (CK-18). Associations between markers of traffic exposure during different age windows and CK-18, stratified by overweight or obese status (orange line represents those who are overweight or obese; purple line represents those who are not overweight or obese). Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. No results were statistically significant after adjustment for multiple comparisons.



eFigure 26: Model results where smooth of exposure had statistically significant interaction with either sex or overweight or obese status for a liver enzyme biomarker outcome. Associations between air pollution or traffic exposures and liver enzyme biomarkers (alanine aminotransferase [ALT] and aspartate aminotransferase [AST]), with exposure smooths stratified by sex or overweight or obese status. Modeled using generalized additive models, adjusting for cohort, maternal age, maternal pre-pregnancy BMI, maternal education level, paternal education level, and maternal active smoking during pregnancy. P-values for interaction (p-for-ixn) are not adjustment for multiple comparisons.